

Take picture of board.

01/11/2025.

Probability

1. • Definitions.

9:35-9:35



$\Omega = \{1, 2, 3, \dots, 6\}$: sample space.

$E = \{1\}, \{2, 4\}, \dots$: event.



$\Omega = \{r \mid 0 \leq r \leq r_{\max}\}$: sample space

$E = \{r^{(1)} = 0.23\}, \dots$: event.

$\{r^{(1)} = 0.1, r^{(2)} = 0.2\}$

⇒ Probability : Event / Sample space.

1) Objective probabilities

$$p = \lim_{N \rightarrow \infty} \frac{N_E}{N}$$

2) Subjective probabilities.

Theoretical estimate. $\sim 1/6$ dice.

Q) Monty Hall problem.

[1]	[2]	[3]	Result (stay #1)	Result (change to offered)
G	G	C	G	C
G	C	G	G	C
C	G	G	C $\rightarrow 1/3$	G $\rightarrow 2/3$

Rule : Host must,

- 1) open door not selected by contestant.
- 2) open door to reveal goat not car.
- 3) offer chance to switch btw original and remaining closed.

2. • Q)

20 people at least two have same birthday

9:35-9:40

$$p = 1 - \frac{365 P_{20}}{365^{20}} \approx 0.4114.$$

3. Rules. (in HW)

9:40-9:45

- Additive: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$



- Conditional probability

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \quad \text{Given A.}$$

(Why? Multiplicative! $\Rightarrow \underbrace{P(A)}_{\text{A already happened}} \cdot \underbrace{P(B|A)}_{\text{Then, both A, B happens.}} = P(A \cap B)$)

- Independence.

$P(B|A) = P(B)$: doesn't care A happens.

$$\Rightarrow P(A) \cdot P(B) = P(A \cap B).$$

Random Variables.

• Discrete RV

Event: $\{X=x\}$

$$\langle X \rangle = \sum_x x P(X=x)$$

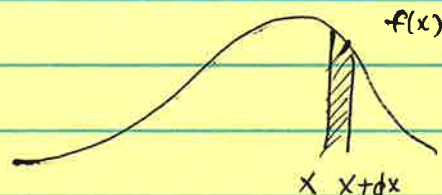
$$\begin{aligned} \text{Var}(X) &= \langle (X - \langle X \rangle)^2 \rangle = \langle X^2 + \langle X \rangle^2 - 2\langle X \rangle X \rangle \\ &= \langle X^2 \rangle + \langle X \rangle^2 - 2\langle X \rangle^2 \\ &= \langle X^2 \rangle - \langle X \rangle^2 \end{aligned}$$

$$\sigma(X) = \{\text{Var}(X)\}^{1/2}$$

$\langle X^k \rangle$ = kth moment.

$\langle g(X) \rangle$ = average.

• Continuous RV



$$f(x) \cdot dx = P\{x \leq X \leq x+dx\}.$$

$$E\{X\} = \int_{-\infty}^{\infty} x f(x) dx.$$

$$\begin{aligned} E\{(X - E\{X\})^2\} &= \text{Var}\{X\}. \\ &= \int_{-\infty}^{\infty} x^2 f(x) dx - \left(\int_{-\infty}^{\infty} x f(x) dx\right)^2. \end{aligned}$$

4. Q) Prove $\langle XY \rangle = \langle X \rangle \langle Y \rangle$ if X, Y are independent

9:45-9:50 A) $\langle XY \rangle = \iint xy \cdot f(x) \cdot f(y) \cdot dx dy = \int y \left(\int x f(x) dx \right) f(y) dy = \langle X \rangle \langle Y \rangle$ #

$$\int x P(X=x) \cdot y P(Y=y) = \int xy P(X=x, Y=y)$$

HW 2-c

• Multi-variate prob. dist.

$$- \langle aX + bY \rangle = a\langle X \rangle + b\langle Y \rangle$$

$$- \text{Cov}(X, Y) = \underbrace{\langle (X - \langle X \rangle) \rangle}_{\text{mean} = 0, \text{Var}(X)} \underbrace{\langle (Y - \langle Y \rangle) \rangle}_{\text{mean} = 0, \text{Var}(Y)} \leadsto \langle \uparrow\uparrow, \uparrow\downarrow, \downarrow\downarrow \rangle$$

$$= \langle XY \rangle - \langle X \rangle \langle Y \rangle \cdot 2 + \langle X \rangle \langle Y \rangle = \langle XY \rangle - \langle X \rangle \langle Y \rangle$$

$$- \text{correlation: } \rho(X, Y) = \text{Cov}(X, Y) / \sigma_X \sigma_Y$$

Q) $X_1 \sim X_N$ where $X_i \sim N(\mu, \sigma^2)$, i.i.d.

$$\text{Define } \bar{X} = \frac{1}{N} \sum_{i=1}^N X_i, \quad \langle \bar{X} \rangle = \mu, \quad \text{Var}(\bar{X}) = \sigma^2/N, \quad \sigma(\bar{X}) = \sigma/\sqrt{N}$$

$$\text{Pf) } \text{Var}(\bar{X}) = \langle (\bar{X} - \langle \bar{X} \rangle)^2 \rangle = \langle \bar{X}^2 \rangle - \langle \bar{X} \rangle^2$$

$$= \frac{1}{N^2} \cdot \langle \underbrace{\quad}_{-\mu^2} \rangle = \frac{1}{N^2} \langle X_1^2 + X_2^2 + \dots + X_N^2 + 2X_1X_2 + \dots + 2X_{N-1}X_N \rangle - \mu^2$$

$$= \frac{1}{N^2} \langle \underbrace{X_1 \cdot \bar{X} \cdot N + X_2 \cdot \bar{X} \cdot N + \dots + X_N \cdot \bar{X} \cdot N}_{\frac{2}{N^2} \mu^2 \cdot \frac{N(N-1)}{2}} \rangle - \mu^2$$

$$= \frac{1}{N^2} \sum_i \langle X_i^2 \rangle + \frac{2}{N^2} \sum_{i \neq j} \langle X_i X_j \rangle - \mu^2$$

$$= \frac{1}{N^2} \sum_i \{ \text{Var}(X_i) + \mu^2 \} + 0 - \mu^2 = \frac{1}{N^2} \cdot N \cdot \text{Var}(X_i) = \sigma^2/N \quad \#$$

5. Central limit Theorem central limit theorem.

6. Fun: Stirling's formula.

$$N! = \sqrt{2\pi N} \cdot (N/e)^N \rightarrow \ln N! = N \ln N - N + \dots$$

discrete.

continuous.

Entropy.

If time permits.

• Gaussian distribution

$$\left(\int_{-\infty}^{+\infty} e^{-x^2} dx \right)^2 = \int_{-\infty}^{+\infty} e^{-x^2} dx \cdot \int_{-\infty}^{+\infty} e^{-y^2} dy = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} e^{-(x^2+y^2)} dx dy$$

$$x = r \cos \theta, \quad y = r \sin \theta.$$

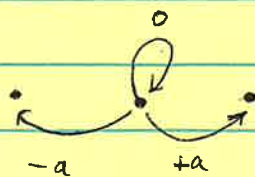
$$\cancel{dx dy} \quad dx dy = r dr d\theta$$

$$= \int_0^{2\pi} \int_0^{\infty} e^{-r^2} \cdot r dr d\theta = 2\pi \cdot \frac{1}{2} = \pi$$

$$\Rightarrow \int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi}.$$

$$\therefore f(x) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\left(\frac{x-\mu}{\sigma}\right)^2\right)$$

• Stationary Random Walk



$$l_i = \begin{cases} +a & \text{prob: } p \\ 0 & \text{prob: } 1-2p \\ -a & \text{prob: } p \end{cases} \quad (p < 1/2) \quad (*)$$

$$\text{Then, } \langle l_i \rangle = a p + 0 \cdot (1-2p) + (-a) p = 0 \quad \text{--- (1)}$$

$$\text{Also, } \langle l_i^2 \rangle = a^2 p + 0^2 (1-2p) + (-a)^2 p = 2a^2 p. \quad \text{--- (2)}$$

we know that $\langle l_i l_j \rangle = \langle l_i \rangle \langle l_j \rangle = 0$ if $i \neq j$ $\because l_i, l_j$ independent.

From the previous work,

$$\begin{aligned} \langle |X(n\tau)|^2 \rangle &= \left\langle \left(\sum_i l_i \right)^2 \right\rangle = \sum_i \langle l_i^2 \rangle + \sum_{i \neq j} \langle l_i l_j \rangle \\ &= n \cdot (2a^2 p) \end{aligned}$$

$$\text{Recall that } \langle |X(t)|^2 \rangle = 2D_s t \cdot d. \quad (d: \text{dimension} = 1.)$$

$$\Rightarrow n \cdot (2a^2 p) = 2D_s n\tau \cdot 1$$

$$\Rightarrow D_s = a^2 p / \tau$$

$$\text{Note that non-stationary case yields } D = \frac{a^2}{2\tau} \quad \text{and} \quad D_s = \frac{a^2}{2\tau} (2p).$$

$$\Rightarrow D_s / D = 2p < 1 \quad (\because *) \quad \therefore D_s < D$$

Q) What is $(2p)$?

$$\text{Hint: Recall } D = \cancel{\mu} k_B T \quad \text{or} \quad D = k_B T / \cancel{\zeta}$$

Problem Session 1

January 10, 2025

Proof of Central Limit Theorem (CLT)

Let $X_1, X_2, \dots, X_N$ be a random sample from an arbitrary distribution with mean μ and variance σ^2 . We shall assume that N is sufficiently large. Define,

$$\bar{X} := \frac{1}{N} \sum_{i=1}^N X_i,$$

where the expectation and the variance of $\bar{X}$ can be calculated as,

$$\langle \bar{X} \rangle = \frac{1}{N} \sum_{i=1}^N \langle X_i \rangle = \mu, \quad \text{Var}(\bar{X}) = \langle (\bar{X} - \langle \bar{X} \rangle)^2 \rangle = \langle \bar{X}^2 \rangle - \langle \bar{X} \rangle^2 = \langle \bar{X}^2 \rangle - \mu^2,$$

Recall that $\langle \bar{X}^2 \rangle$ reads,

$$\langle \bar{X}^2 \rangle = \left\langle \left(\frac{1}{N} \sum_{i=1}^N X_i \right) \left(\frac{1}{N} \sum_{i=1}^N X_i \right) \right\rangle = \frac{1}{N^2} \left(\sum_{i=1}^N \langle X_i^2 \rangle + 2 \sum_{i \neq j} \langle X_i X_j \rangle \right),$$

and because X_i are independent samples, $\langle X_i X_j \rangle = \langle X_i \rangle \langle X_j \rangle = \mu^2$. Also note that $\langle X_i^2 \rangle = \text{Var}(X_i) + \langle X_i \rangle^2 = \sigma^2 + \mu^2$ so that,

$$\langle \bar{X}^2 \rangle = \frac{1}{N^2} \left(N (\sigma^2 + \mu^2) + 2 \binom{N}{2} \mu^2 \right) = \frac{\sigma^2 + \mu^2}{N} + \frac{N-1}{N} \mu^2 = \frac{\sigma^2}{N} + \mu^2,$$

so that the expectation and the variance of $\bar{X}$ is,

$$\langle \bar{X} \rangle = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{N}.$$

Semi-Stationary Random Walk

Let us define a semi-stationary random walk, which is described as a random walker in 1-dimensional space. At each time step, the random walker can move either $+a$ or $-a$ with probability p and also can stay at its state with probability $1 - 2p$. We call this random walk as semi-stationary because the random walker can stay where it is.

Define a step that the random walker takes at i^{th} time step to be l_i . Then, the expectation of l_i and l_i^2 can be calculated as,

$$\langle l_i \rangle = (+a)p + (0)(1 - 2p) + (-a)p = 0, \quad \langle l_i^2 \rangle = (a^2)p + 0^2(1 - 2p) + (a^2)p = 2a^2p,$$

which can be used to calculate $\langle |X(n\tau)|^2 \rangle$ as,

$$\langle |X(n\tau)|^2 \rangle = \left\langle \left(\sum_i l_i \right)^2 \right\rangle = \sum_i \langle l_i^2 \rangle = n(2a^2p),$$

assuming that l_i and l_j are independent when $i \neq j$. Recall that $\langle |X(n\tau)|^2 \rangle = 2D_s n\tau$,

$$2D_s n\tau = n(2a^2p), \quad D_s = 2pD$$

where $D = a^2/(2\tau)$ and D_s is a diffusion coefficient for the semi-stationary random walk. Hence, we have derived that the probability of moving in the semi-stationary random walk drives the mobility of the diffusion. In other words, the action of staying at its state works as a friction which slows down the diffusion.

Problem Session 2

01/17/2025

Lagrangian. **09:30 - 09:40**

$$F_i = m \cdot \ddot{q}_i \Leftrightarrow d p_i / dt = F_i \Leftrightarrow \ddot{q}_i = -\frac{1}{m} \cdot \partial U / \partial q_i \quad (\text{Newton, 1687})$$

Define,

$$L(\{q_i\}, \{\dot{q}_i\}) = K - U = \sum_i \left(\frac{1}{2} m \dot{q}_i^2 - U(q_i) \right) \quad \text{where,}$$

$$\frac{d}{dt} \left(\partial L / \partial \dot{q}_i \right) - \left(\partial L / \partial q_i \right) = 0 \quad \text{for } \forall i \quad (\text{Lagrange, 1760})$$

$$p_i) \quad \partial L / \partial \dot{q}_i = m \cdot \dot{q}_i \equiv p_i$$

$$\partial L / \partial q_i = - \partial U / \partial q_i$$

$$\Rightarrow \frac{d}{dt} p_i = - \partial U / \partial q_i = F_i$$

Using derivatives,

$$dL = \sum_i \frac{\partial L}{\partial q_i} dq_i + \frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i$$

$$\Rightarrow \frac{dL}{dt} = \sum_i \underbrace{\frac{\partial L}{\partial q_i} \frac{dq_i}{dt}}_{= ①} + \underbrace{\frac{\partial L}{\partial \dot{q}_i} \frac{d\dot{q}_i}{dt}}_{= \frac{\partial L}{\partial \dot{q}_i} \frac{d}{dt}(\dot{q}_i)}$$

$$\begin{aligned} ① &= \frac{\partial L}{\partial q_i} \frac{dq_i}{dt} = \left(\partial L / \partial q_i \right) \cdot \left(\partial \dot{q}_i / \partial q_i \right) \cdot \underbrace{\frac{dq_i}{dt}}_{= \dot{q}_i} = \underbrace{\left(\partial \dot{q}_i / \partial q_i \right)}_{= d/dt} \left(\partial L / \partial \dot{q}_i \right) \cdot \dot{q}_i \\ &= \frac{d}{dt} \left(\partial L / \partial \dot{q}_i \right) \dot{q}_i \end{aligned}$$

$$\Rightarrow dL/dt = \sum_i \frac{d}{dt} \left(\partial L / \partial \dot{q}_i \right) \dot{q}_i + \partial L / \partial q_i \frac{d}{dt}(\dot{q}_i) = \sum_i \frac{d}{dt} \cdot \left(\partial L / \partial \dot{q}_i \right) \dot{q}_i$$

$$\therefore \frac{d}{dt} \left[L - \sum_i \frac{\partial L}{\partial \dot{q}_i} \dot{q}_i \right] = 0$$

Hamiltonian 09:40 - 09:50

Define $H = -L + \sum_i \left(\partial L / \partial \dot{q}_i \right) \dot{q}_i \Rightarrow \frac{d}{dt}(H) = 0 \Rightarrow H$ is conserved.

Legendre's Transform. (1787) $\langle L \rightarrow H \rangle$

Note: $H = -L + \sum_i \left(\partial L / \partial \dot{q}_i \right) \dot{q}_i$ and $\partial L / \partial \dot{q}_i = m \cdot \dot{q}_i \equiv p_i$ (*)

$$\Rightarrow H = -L + \sum_i p_i \dot{q}_i$$

Also, $\partial L / \partial \dot{q}_i = \frac{d}{dt} \left(\partial L / \partial \dot{q}_i \right) = \dot{p}_i \Rightarrow$ Force?!

$$\therefore \partial L / \partial \dot{q}_i \equiv p_i$$

$$\partial L / \partial \dot{q}_i \equiv \dot{p}_i$$

$$dH = \sum_i p_i d\dot{q}_i + \dot{q}_i dp_i - \frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i - \frac{\partial L}{\partial p_i} dp_i$$

$= \dot{p}_i d\dot{q}_i - \dot{q}_i dp_i$

Key point: Legendre Transform eliminates the dependence of the function L respect to the variable $\dot{q}_i$.

Example)

① $H(p, x) = \frac{p^2}{2m} + V(x)$: Function of p, x .

I want to replace "p" with something else!

$$\Rightarrow L = H - \left(\frac{\partial H}{\partial p} \right)_x p = H - v \cdot p \quad \left(\because \frac{\partial}{\partial p} \left(\frac{p^2}{2m} \right) = p/m = v \right)$$

$$\Rightarrow L = H - v \cdot p \quad : \text{constant for } (x) \text{ and } (v)$$

check: ① $dH = \left(\frac{\partial H}{\partial p} \right)_x dp + \left(\frac{\partial H}{\partial x} \right)_p dx = (v)_x dp + (V'(x))_p dx$: Function of (p, x)

$$\textcircled{2} dL = dH - v dp - p dv = v dp + V'(x) dx - v dp - p dv$$

$$= V'(x) dx - p dv$$

Function of (v, x)

Note: From (*)

$$dH = \sum_i \dot{p}_i d\dot{q}_i + \dot{q}_i dp_i$$

$$dH = \sum_i \frac{\partial H}{\partial \dot{q}_i} d\dot{q}_i + \frac{\partial H}{\partial p_i} dp_i$$

< Eq. motion Hamilt. >

$$\dot{p}_i = -\partial H / \partial \dot{q}_i$$

$$\dot{q}_i = \partial H / \partial p_i$$

Example on Thermodynamics.

$$E = TS - PV + \mu N \quad \text{and} \quad dE = Tds - pdV + \mu dN$$

$$E(s, V, N)$$

① $E - \left(\frac{\partial E}{\partial s} \right)_{V, N} s \equiv A$ (Helmholtz)

$$dA = dE - Tds - sdT = \cancel{Tds} - \cancel{pdV} + \cancel{\mu dN} - \cancel{Tds} - \cancel{sdT} \quad [N, V, T]$$

② $E - TS - \left(\frac{\partial E}{\partial V} \right)_{S, N} V \equiv G$ (Gibbs)

$$dG = \cancel{Tds} - \cancel{pdV} + \cancel{\mu dN} + \cancel{pdV} - \cancel{Tds} - \cancel{sdT} + Vdp \quad [N, P, T]$$

Example on pendulum. 09:40 - 09:50

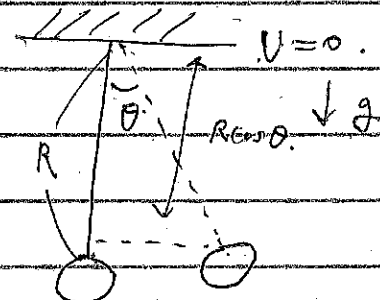
① $K = \frac{1}{2} m \cdot v^2$
 $= \frac{1}{2} m (R \dot{\theta})^2 = \frac{1}{2} m R^2 (\dot{\theta})^2$

$$U = -mgR \cos \theta$$

Formulate Lagrangian as,

$$\Rightarrow \mathcal{L} = K - U = \frac{1}{2} m R^2 (\dot{\theta})^2 + mgR \cos \theta$$

function of $\theta, \dot{\theta}$



② Equation of motion.

1) Lagrangian

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) - \frac{\partial \mathcal{L}}{\partial \theta} = 0$$

$$\Rightarrow \frac{d}{dt} (m R^2 \dot{\theta}) - (-mgR \sin \theta) = 0$$

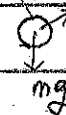
$$\Rightarrow \ddot{\theta} = -g/R \sin \theta$$

2) Newtonian

$$T = mg \cos \theta$$

$$T - mg \sin \theta = m \cdot R \ddot{\theta}$$

$$\Rightarrow \ddot{\theta} = -g/R \sin \theta$$



③ To Hamiltonian, (by Legendre Transform).

$L(\theta, \dot{\theta}) \rightarrow$ we want to eliminate $\dot{\theta}$

$$H^* = -L + \left(\frac{\partial L}{\partial \dot{\theta}} \right) \dot{\theta} \quad \text{where} \quad p_{\theta} = \frac{\partial L}{\partial \dot{\theta}} = mR^2 \dot{\theta}$$

$$= -\frac{1}{2} mR^2 (\dot{\theta})^2 + mgR \cos \theta + mR^2 \dot{\theta} \cdot \dot{\theta}$$

$$= -mgR \cos \theta + \frac{p_{\theta}^2}{(2mR^2)} \quad (\text{sign flip of } \cos)$$

$$H^* = \frac{p_{\theta}^2}{(2mR^2)} - mgR \cos \theta.$$

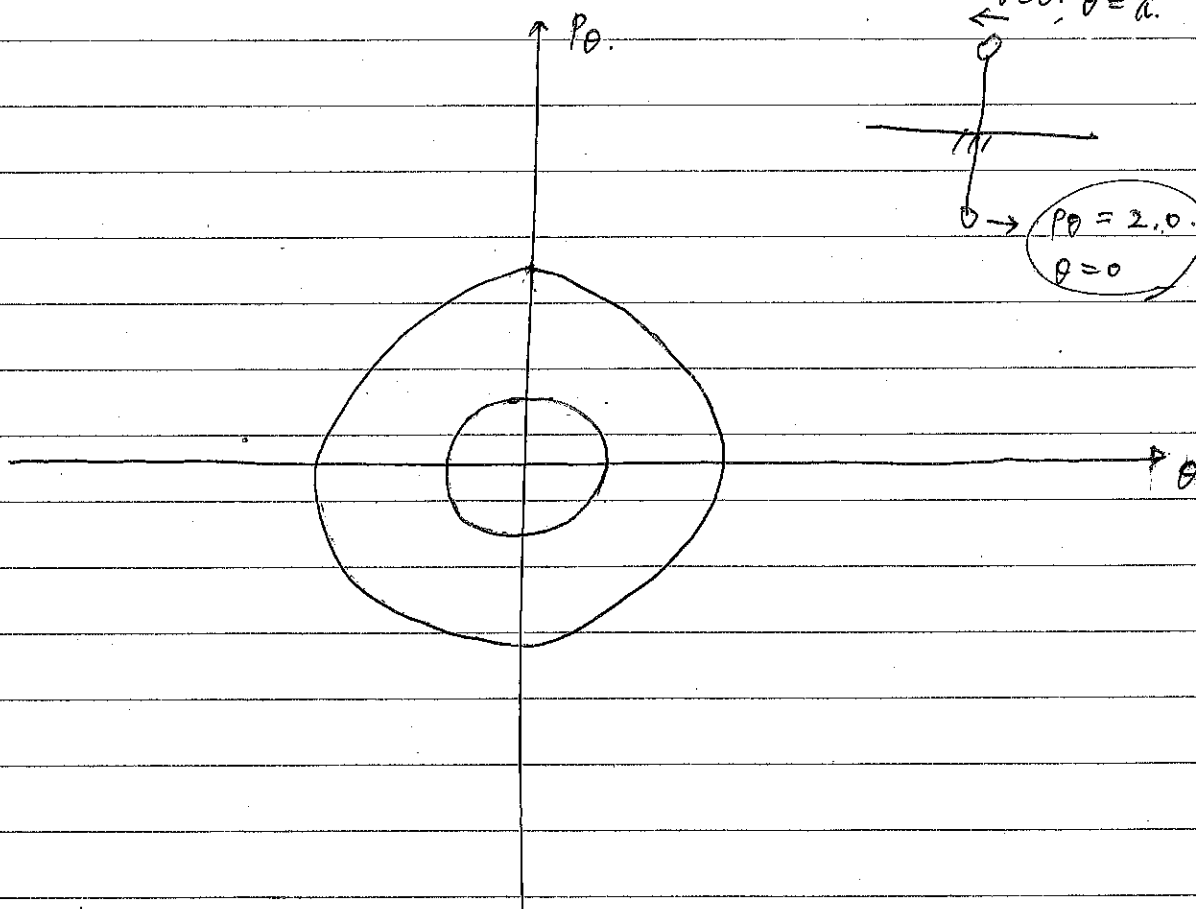
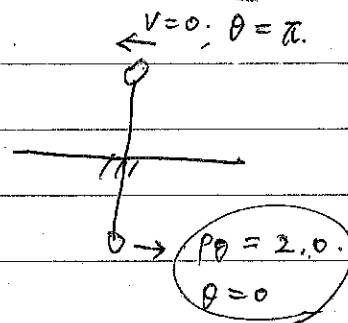
Recall eq. mot. Hamilt.

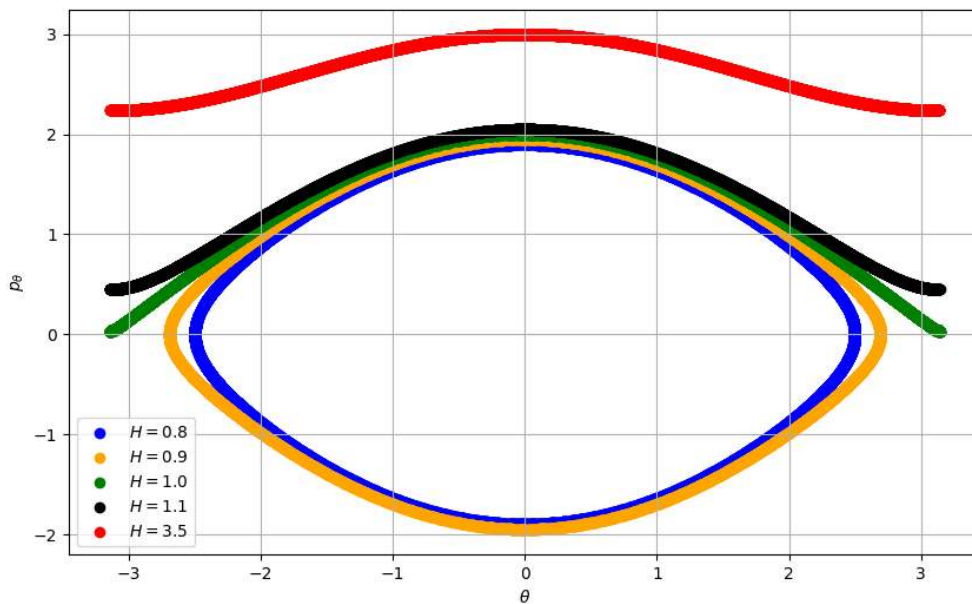
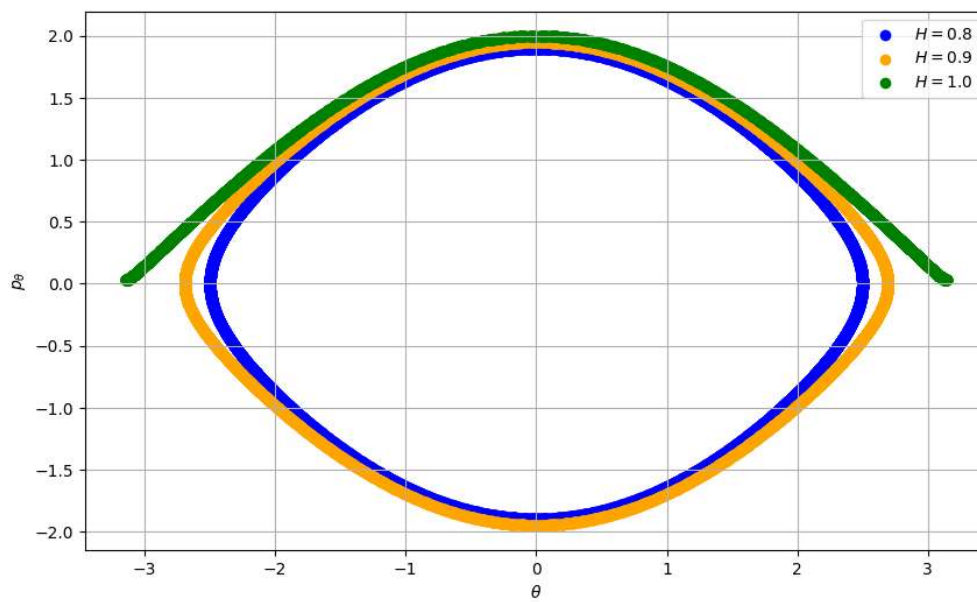
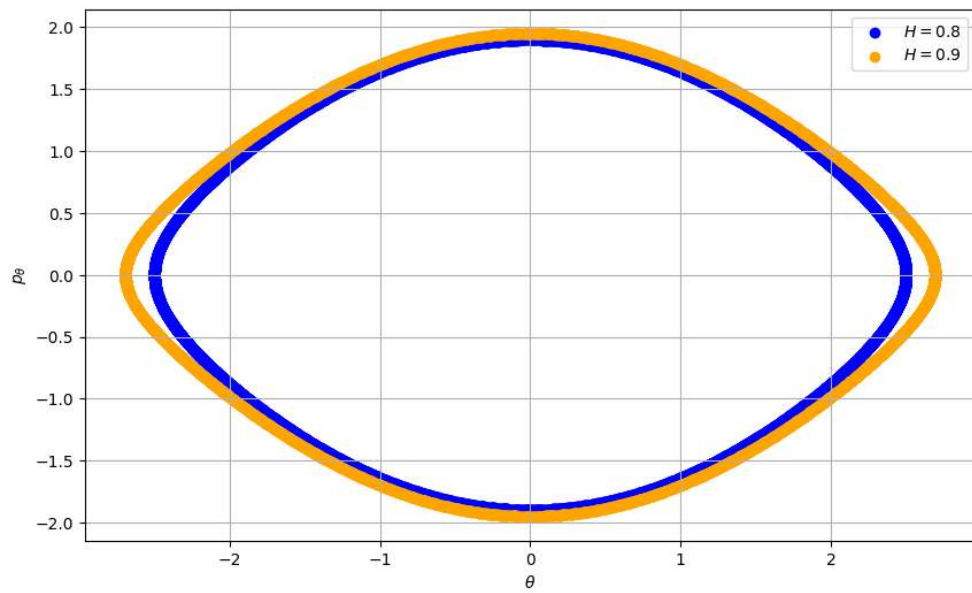
09:50 - 10:00

$$\dot{q}_i = -\partial H / \partial p_i = -(-mgR \cdot (-\sin \theta)) = -mgR \sin \theta.$$

$$\dot{p}_i = +\partial H / \partial q_i = p_{\theta} / (mR^2)$$

$H = 1$





ME 346A Problem Session. (01/31/2025)

- Properties of Ideal Gas. (p. 24-27 of notes), ch. 6.

09:30-09:40

$$S = k_B \cdot N \cdot \left[\frac{5}{2} + \log \left(\frac{V}{N} \cdot \left(\frac{4\pi m E}{3N h^2} \right)^{3/2} \right) \right] \quad \left(\text{Refer p. 10 Ch 7 for derivation with hypersphere assumption} \right)$$

09:30-09:35 (a)

$$E(S, V, N) = ? , T = ? , p = ? , \mu = ?$$

Does $pV = Nk_B T$ hold?

$$E(S, V, N) = \frac{3N h^2}{4\pi m} \left(\frac{N}{V} \right)^{2/3} \exp \left[\frac{2S}{3Nk_B} - \frac{5}{3} \right]$$

$V^{-2/3}$

$$\textcircled{1} \quad T = \left(\partial E / \partial S \right)_{V, N} = \frac{2}{3Nk_B} \cdot E(S, V, N) \Rightarrow E = \frac{3}{2} Nk_B T.$$

$$\textcircled{2} \quad p = - \left(\partial E / \partial V \right)_{S, N} = - \left(-2/3 \cdot V^{-5/3} \cdot \textcircled{1} \right) = \frac{2}{3} \cdot \frac{1}{V} E \\ = Nk_B T / V \Rightarrow pV = Nk_B T.$$

$$\textcircled{3} \quad \mu = \left(\partial E / \partial N \right)_{S, V} = E/N \cdot \left(5/3 - 2S/(3Nk_B) \right)$$

09:40-09:50

09:35-09:40 (b)

$$A = E - TS = \frac{3}{2} Nk_B T - T Nk_B \cdot \left[\log \left(\frac{V}{N} \cdot \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2} \right) + 5/2 \right]$$

(c)

$$G = E - TS + pV = A + pV$$

$$= -Nk_B T \left[\log \left(\frac{k_B T}{p} \cdot \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2} \right) \right]$$

From $\textcircled{3}$, of (a),

$$\mu N = E \left(5/3 - 2S/(3Nk_B) \right) = \frac{5}{2} Nk_B T - TS$$

$$= \textcircled{G}$$

$$\Rightarrow G = \mu N.$$

09:50-10:00

09:40-09:45 (d) $C_V = \left(\frac{\partial Q}{\partial T} \right)_{V,N} = T \cdot \left(\frac{\partial S}{\partial T} \right)_{V,N} = -T \cdot \left(\frac{\partial^2 A}{\partial T^2} \right)_{V,N}$

since $S = - \left(\frac{\partial A}{\partial T} \right)_{N,V}$

$\therefore A = E - TS$

$\Rightarrow C_V = T \left(\frac{\partial S}{\partial T} \right) = \frac{3}{2} N k_B$

09:45-09:50 (e) $C_P = \left(\frac{\partial Q}{\partial T} \right)_{P,N} = T \left(\frac{\partial S}{\partial T} \right)_{P,N} = -T \left(\frac{\partial^2 G}{\partial T^2} \right)_{P,N}$

since $S = - \left(\frac{\partial G}{\partial T} \right)_{P,N}$

$\therefore G = E - TS + pV$

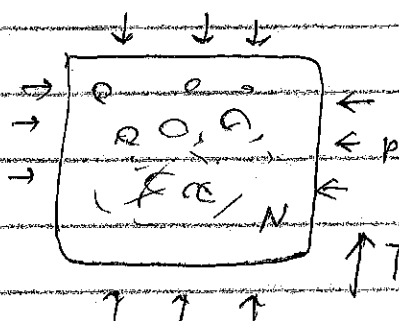
$\Rightarrow C_P = \frac{5}{2} N k_B$

$\Rightarrow C_P - C_V = N k_B$

10:00-10:10

09:45-09:50 (e) At constant pressure, and fixed N , we want to know

$\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_{P,N}$



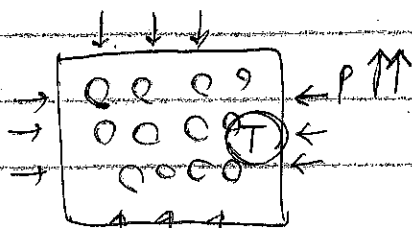
From $pV = N k_B T$, $V = N k_B T / p$

$\Rightarrow \alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_{N,p} = \frac{p}{N k_B T} \cdot \frac{N k_B}{p} = 1/T$

$\Rightarrow \alpha = 1/T$

At constant temperature, and fixed N , we want to know

$\beta = - \frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_{T,N}$



$\beta = + \frac{1}{V} \cdot \left(\frac{N k_B T}{p^2} \right) = 1/p$

$\Rightarrow \beta = 1/p$

$\alpha^2 V T / p = \frac{1}{T^2} V T p$
 $= N k_B = C_P - C_V$

10:10-10:20

Notes on Legendre Transform & Laplace Transform.

$$Z(\beta) = \sum_{N|E(N)=E} e^{-\beta E(N)} \approx \int dE \cdot \Omega(E) \cdot e^{-\beta E(N)} \quad : \text{Laplace Transform.} \quad \text{①}$$

$$\text{①} = \int dE \cdot \Omega(E) e^{-\beta E(N)} = \int dE \cdot \exp \left[-N \left(\beta E - \log \Omega(E) \right) \right] \\ = A(E) \\ = A(E)$$

$$\text{where } A = E - T \cdot S = E - \left(\frac{\partial E}{\partial S} \right)_{N,V} \cdot S \quad : \text{Legendre Transform} \quad \text{②}$$

① & ② are related.

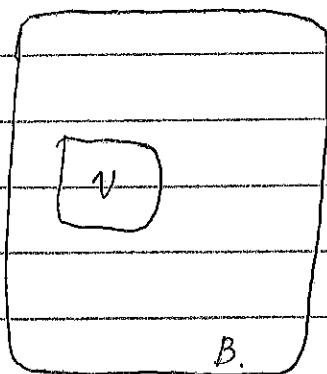
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ME346A

Problem Session. (01/31/2025).

01/27/2025.

- Micro-canonical Ensemble.



$$E_T = E(N) + E_B.$$

↗ partition function.

$$p(N, B) = 1 / \Omega(N_T, V_T, E_T)$$

$$\Rightarrow p(N) = \sum_{\{B \mid E(N) + E(B) = E(T)\}} p(N, B) = \frac{1}{\Omega_T} \cdot \Omega_B.$$

$$\Rightarrow p(N) = \frac{1}{\Omega_T} \cdot \Omega_B(E_T - E(N), N_B, V_B)$$

$N_T, V_T, E_T \rightarrow \text{Fixed.}$

$$\log p(N) \propto \log \Omega_B(E_T - E(N), N_B, V_B) = \log \Omega_B(E_T)$$

$$- \left(\frac{\partial \log \Omega_B(N_B, V_B, E_B)}{\partial E_B} \right) \cdot E(N)$$

$$+ O((E(N))^2)$$

$$\Rightarrow \log p(N) \propto - \underbrace{\left(\frac{\partial \log \Omega_B}{\partial E_B} \right)_{N_B, V_B}}_{(=\beta)} E(N).$$

(=β)

$$\Rightarrow p(N) \propto e^{-\beta E(N)}$$

! Boltzmann Distribution!

$$\text{Z}(\beta) = \sum_N e^{-\beta E(N)} = \sum_E \Omega(E) \cdot e^{-\beta E} \quad \langle \text{Laplace} \rangle$$

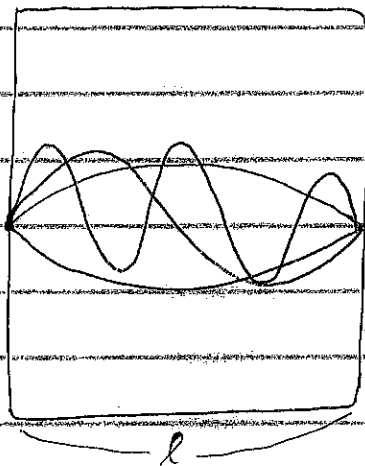
$$\Rightarrow \text{Z}(\beta) \approx \int dE \cdot \Omega(E) e^{-\beta E} = \int dE \exp(-\beta [E - \beta^{-1} \log \Omega(E)])$$

$$= \int dE \exp(-N\beta \cdot (\underbrace{\epsilon - T_s}_{=a}))$$

< Legendre >

= a

• Ideal Gas Law.



→ Translational only,

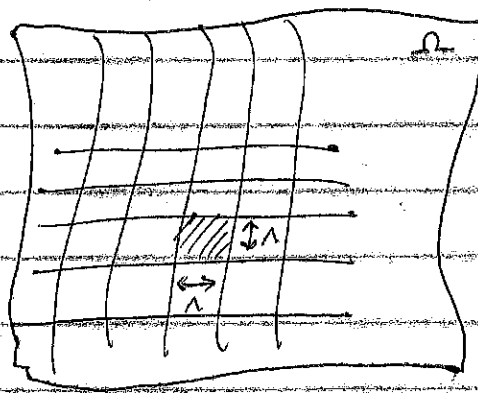
$$\text{Quantum Mechanics} = \frac{n^2 h^2}{8 m l^2} \equiv \epsilon_n \quad (\text{quantized})$$

$$Z(\beta) = \sum_{n=1}^{\infty} e^{-\beta \epsilon_n}$$

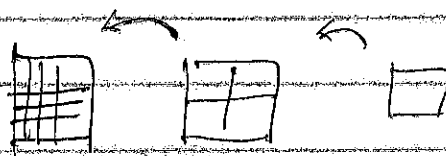
$$Z(\beta) \approx \int_0^{\infty} e^{-\beta \epsilon_n} dn = \frac{\sqrt{2\pi m k_B T}}{h} \quad (\beta \ll 1) \quad (2)$$

Extend (2) to 3D, $\epsilon_{n,3D} = \frac{h^2}{8m} \left[\left(\frac{n_x}{l_x} \right)^2 + \left(\frac{n_y}{l_y} \right)^2 + \left(\frac{n_z}{l_z} \right)^2 \right]$

$\Rightarrow \underbrace{Z(\beta)_{3D}}_{\text{small } Z} = \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2} \cdot l_x l_y l_z = \frac{V}{\Lambda^3} \quad \left(\Lambda = \left(\frac{2\pi m k_B T}{h^2} \right)^{-1/2} \right)$



Λ : Thermal De Broglie wavelength.



$$A = -\beta^{-1} \log Z(\beta) \quad , \quad A = E - TS \quad , \quad Z(\beta) = \frac{1}{N!} (Z(\beta))^N$$

$$\Rightarrow -p = \left. \frac{\partial A}{\partial V} \right|_{N,T} = -\beta^{-1} \cdot \frac{\partial}{\partial V} \left(\frac{1}{N!} \log \left(\frac{V}{\Lambda^3} \right)^N \right) = -\beta^{-1} \cdot \frac{N}{V}$$

$$\therefore pV = N \cdot k_B T \quad \Leftrightarrow \quad pP = p$$

- From Quantum Mechanics Partition Function.

$$Z(\beta) = 1/N! z^N(\beta) \quad (\text{where } z(\beta) = V/\lambda^3)$$

$$\langle E \rangle = \frac{\partial}{\partial(-\beta)} \log Z(\beta) = Nk_B T^2 \left(\frac{\partial}{\partial T} \log z(\beta) \right)_{N,V}$$

$$= Nk_B T^2 \cdot \frac{\frac{3}{2} \cdot \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2} \cdot \frac{2\pi m k_B}{h^2}}{\left(\frac{2\pi m k_B T}{h^2} \right)^{3/2}} = \frac{3}{2} Nk_B T^2 \frac{1/\lambda}{1/\lambda^3} \frac{1}{T}$$

$$\Rightarrow \boxed{\langle E \rangle = \frac{3}{2} Nk_B T.} \quad \text{as } N \rightarrow \infty, \quad E = \frac{3}{2} Nk_B T \quad (\because \text{var}(E) \downarrow 0)$$

- Some Thermo - properties.

$$p = - \left(\frac{\partial E}{\partial V} \right)_{S,N} = Nk_B T/V$$

$$\mu = \left(\frac{\partial E}{\partial N} \right)_{S,V} = \frac{3}{2} k_B T.$$

$$\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p = 1/T, \quad \gamma = - \frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_{T,N} = 1/p.$$

$\Rightarrow$ Similar to Sackur - Tetrode equation.

Problem Session 02/07/2025.

09:30 → 09:40

Ideal Gas.

$$Z = \frac{1}{N!} \frac{1}{h^{3N}} \int \prod_{i=1}^{3N} dq_i dp_i \exp\left(-\frac{H(\{q_i\}, \{p_i\})}{k_B T}\right)$$

$$H = \sum_i \frac{p_i^2}{2m} + V(\{q_i\})$$

$$\Rightarrow Z = \frac{1}{N!} \frac{1}{h^{3N}} \int \prod_{i=1}^{3N} dq_i dp_i \exp\left(-\frac{p^2}{2m k_B T}\right)$$

$$= \frac{1}{N!} \frac{1}{h^{3N}} \cdot V \cdot \left[\int_{-\infty}^{\infty} dp \cdot \exp\left(-\frac{p^2}{2m k_B T}\right) \right]^{3N}$$

$$= \frac{V^N}{N! h^{3N}} (2\pi m k_B T)^{3N/2}$$

$$\therefore Z = \frac{V^N}{N! h^{3N}} (2\pi m k_B T)^{3N/2} \quad \text{--- ①}$$

$$\therefore A = -k_B T \ln Z = -k_B T \cdot N \left[\ln\left(\frac{V}{N} \left(\frac{2\pi m k_B T}{h^2}\right)^{3/2}\right) + 1 \right] \quad \#$$

09:40 → 09:45

Quantum (if time allows).

$$\epsilon_n = \frac{n^2 h^2}{8mL^2} \Rightarrow Z = \sum_{n=1}^{\infty} e^{-\beta \epsilon_n} = \frac{\sqrt{2\pi m k_B T}}{h} \quad (\beta \epsilon_1 \ll 1)$$

$$\text{In 3D, } Z = \left(\frac{2\pi m k_B T}{h^2}\right)^{3/2} \cdot dx dy dz = V/\Lambda^3$$

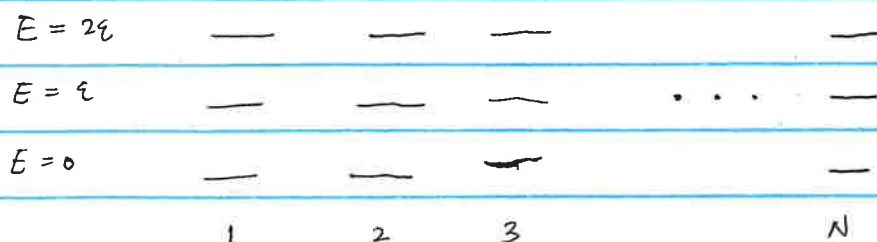
$$\text{With } N \text{ particles, } Z^N \cdot \frac{1}{N!} = \frac{V^N}{N! h^{3N}} (2\pi m k_B T)^{3N/2} \quad \text{--- ②}$$

$\therefore$ ② recovers ①,

At high temp limit, classical $\equiv$ quantum.

09:45 → 09:55

Molecules. with 3 energy.



~~$$Z = \sum_{i=1}^N e^{-\beta H_i}$$~~

$$Z = \sum_{\{n_i\}} e^{-\beta H} \quad H = \sum_{i=1}^N \varepsilon n_i$$

$$1-1) \quad e^{-\beta H} = \exp\left(-\beta \cdot \varepsilon \cdot \sum_{i=1}^N n_i\right) = \exp(-\beta \varepsilon n_1) \cdot \exp(-\beta \varepsilon n_2) \dots$$

$$= \prod_{i=1}^N e^{-\beta \varepsilon n_i}$$

$$1-2) \quad \Rightarrow Z = \sum_{\{n_i\}} \prod_{i=1}^N e^{-\beta \varepsilon n_i} = \prod_{i=1}^N \left(\sum_{n_i=0,1,2} e^{-\beta \varepsilon n_i} \right)$$

$$= \left(1 + e^{-\beta \varepsilon} + e^{-2\beta \varepsilon} \right)^N$$

$$2) \quad A = -k_B T \ln Z = -N k_B T \ln(1 + e^{-\beta \varepsilon} + e^{-2\beta \varepsilon})$$

$$S = -\partial A / \partial T = -N k_B \ln(1 + e^{-\beta \varepsilon} + e^{-2\beta \varepsilon}) - N k_B T \cdot \frac{(-\varepsilon e^{-\beta \varepsilon} - 2\varepsilon e^{-2\beta \varepsilon})}{1 + e^{-\beta \varepsilon} + e^{-2\beta \varepsilon}} \left(-\frac{1}{k_B T^2} \right)$$

$$\Rightarrow S = N k_B \ln(1 + e^{-\beta \varepsilon} + e^{-2\beta \varepsilon}) + N k_B T \cdot \frac{1}{k_B T^2} \cdot \frac{\varepsilon e^{-\beta \varepsilon} + 2\varepsilon e^{-2\beta \varepsilon}}{1 + e^{-\beta \varepsilon} + e^{-2\beta \varepsilon}}$$

$$= N/T$$

$$E = A + TS = N \frac{\varepsilon \cdot e^{-\beta \varepsilon} + 2\varepsilon e^{-2\beta \varepsilon}}{1 + e^{-\beta \varepsilon} + e^{-2\beta \varepsilon}}$$

average expectation

09:55 → 10:03) $\langle E \rangle = - \frac{\partial}{\partial \beta} \ln Z = \frac{\partial \log(Z)}{\partial(-\beta)} = \frac{\partial}{\partial(-\beta)} \cdot N \cdot \log(1 + e^{-\beta \epsilon} + e^{-2\beta \epsilon})$

$$= \frac{\epsilon e^{-\beta \epsilon} + 2\epsilon e^{-2\beta \epsilon}}{1 + e^{-\beta \epsilon} + e^{-2\beta \epsilon}} \cdot N$$

~ Expectation.

$$= N \cdot \epsilon \cdot \left(\frac{e^{-\beta \epsilon} + 2e^{-2\beta \epsilon}}{1 + e^{-\beta \epsilon} + e^{-2\beta \epsilon}} \right)$$

10:00 → 10:05 4) Heat capacity,

$$C_V = \left(\frac{\partial E}{\partial T} \right)_N$$

$$= \frac{\partial \beta}{\partial T} \cdot \frac{\partial E}{\partial \beta} = - \frac{1}{k_B T^2} \frac{\partial E}{\partial \beta} = \frac{1}{k_B T^2} \cdot \frac{\partial E}{\partial(-\beta)}$$

$$= \frac{1}{k_B T^2} \cdot N \cdot \epsilon \cdot \left[\frac{(\epsilon e^{-\beta \epsilon} + 4\epsilon e^{-2\beta \epsilon})(1 + e^{-\beta \epsilon} + e^{-2\beta \epsilon}) - (e^{-\beta \epsilon} + 2e^{-2\beta \epsilon})(\epsilon)(e^{-\beta \epsilon} + 2e^{-2\beta \epsilon})}{(1 + e^{-\beta \epsilon} + e^{-2\beta \epsilon})^2} \right]$$

$$= \frac{1}{k_B T^2} \cdot N \cdot \left(\left[\frac{\epsilon^2 e^{-\beta \epsilon} + 4\epsilon^2 e^{-2\beta \epsilon}}{1 + e^{-\beta \epsilon} + e^{-2\beta \epsilon}} - \left(\frac{\epsilon e^{-\beta \epsilon} + 2\epsilon e^{-2\beta \epsilon}}{1 + e^{-\beta \epsilon} + e^{-2\beta \epsilon}} \right)^2 \right] \right)$$

$$= \underbrace{\text{Var}(E)}_{\text{Var}(E)} \langle E^2 \rangle$$

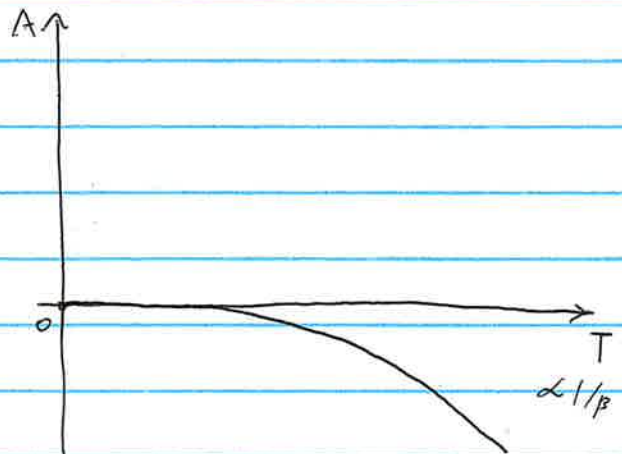
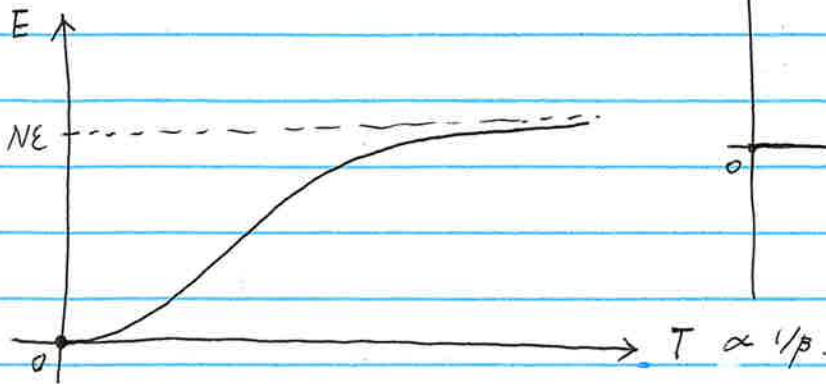
$$\langle E \rangle^2$$

$$= \langle E^2 \rangle - \langle E \rangle^2 = \underbrace{\text{Var}(E)}_{\text{Var}(E)}$$

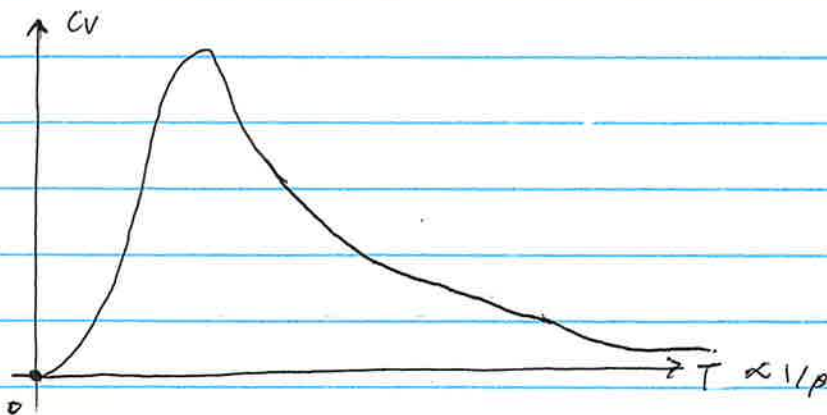
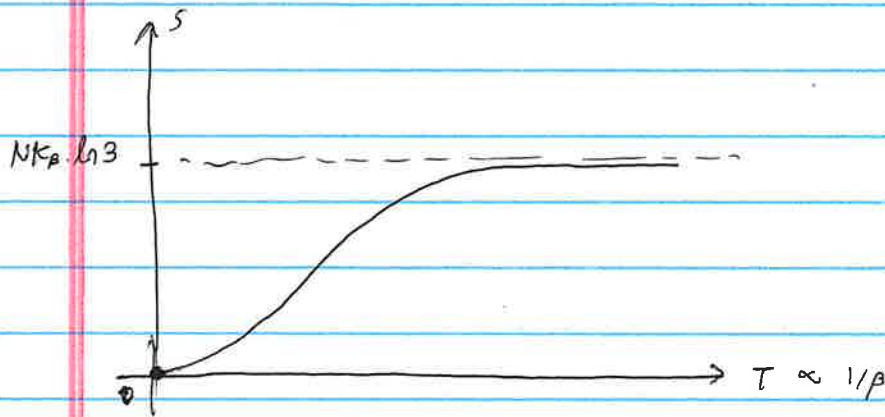
$$A = -Nk_B T \cdot \ln(1 + e^{-\beta \epsilon} + e^{-2\beta \epsilon})$$

10:05 → 10:15 5)

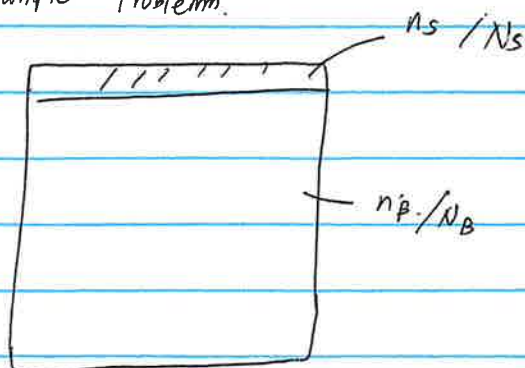
$$E = N \left(\frac{\epsilon \cdot e^{-\beta \epsilon} + 2\epsilon e^{-2\beta \epsilon}}{1 + e^{-\beta \epsilon} + e^{-2\beta \epsilon}} \right)$$



$$S = Nk_B \ln(1 + e^{-\beta \epsilon} + e^{-2\beta \epsilon}) + \underbrace{\frac{N \cdot \frac{1}{\beta}}{k_B}}_{= N/T} \cdot \frac{\epsilon e^{-\beta \epsilon} + 2\epsilon e^{-2\beta \epsilon}}{1 + e^{-\beta \epsilon} + e^{-2\beta \epsilon}}$$



10:15 → 10:30 ° Example Problem.



$$n = n_s + n_B.$$

(a) $E_s = -n_s \cdot \epsilon$

$$S_s = k_B \cdot \ln \left(\frac{N_s!}{n_s! (N_s - n_s)!} \right)$$

$$A_s = E_s - T \cdot S_s = -n_s \cdot \epsilon - T \cdot k_B \cdot \ln \left(\frac{N_s!}{n_s!} \right)$$

(b) $E_B = -\cancel{n_B} \cdot \epsilon$ °

$$S_B = k_B \cdot \ln \left(\frac{N_B!}{n_B! (N_B - n_B)!} \right)$$

$$A_B = E_B - T \cdot S_B = -\cancel{n_B} \cdot \epsilon - T \cdot k_B \cdot \ln \left(\frac{N_B!}{n_B!} \right)$$

(c) Using Stirling's formula,

$$A_s \approx -n_s \cdot \epsilon - k_B T \cdot \left[N_s \ln N_s - n_s \ln n_s - (N_s - n_s) \ln (N_s - n_s) \right]$$

$$A_B \approx 0 - k_B T \cdot \left[N_B \ln N_B - n_B \ln n_B - (N_B - n_B) \ln (N_B - n_B) \right]$$

$$\frac{\partial A}{\partial n_s} = -\epsilon + k_B T \left[\ln n_s + 1 - \ln (N_s - n_s) - 1 \right] - k_B T \left[\ln n_B + 1 - \ln (N_B - n_B) - 1 \right] \quad (\because n - n_s = n_B)$$

$A = A_s + A_B$

$$= -\epsilon - k_B T \cdot \ln \left[\frac{(N_s - n_s)(n - n_s)}{n_s (N_B - n_B)} \right]$$

Using, $N_s, N_B \gg n_s, n_B \Rightarrow \frac{\partial A}{\partial n_s} \approx -\epsilon - k_B T \cdot \ln \left[\frac{(n - n_s) N_s}{n_s \cdot N_B} \right] = 0$

(d) In (c), reads,

$$n_s = \frac{N_s \cdot n}{N_p \cdot \exp(-\beta \epsilon) + N_s} \quad (*)$$

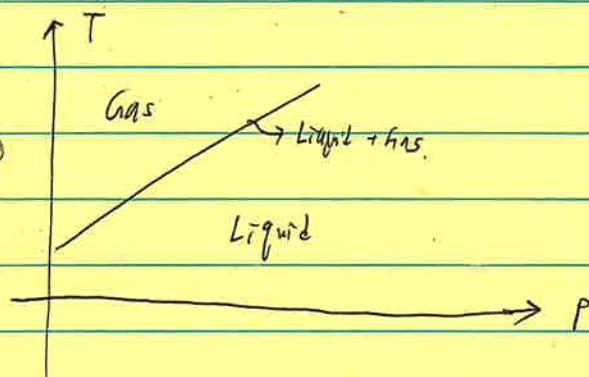
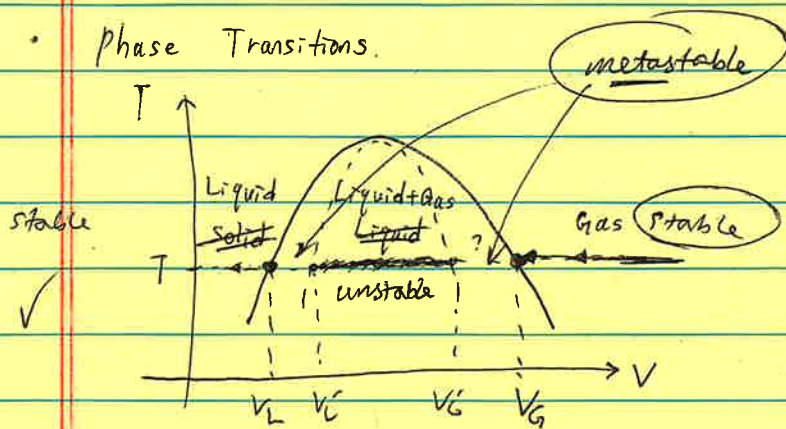
$$\text{Also, } \partial A_s / \partial n_s = -\epsilon - k_B T \cdot \ln \left(\frac{N_s - n_s}{n_s} \right) = \mu_s$$

$$\partial A_B / \partial n_B = k_B T \ln \left(\frac{N_B - n_B}{n_B} \right) = \mu_B.$$

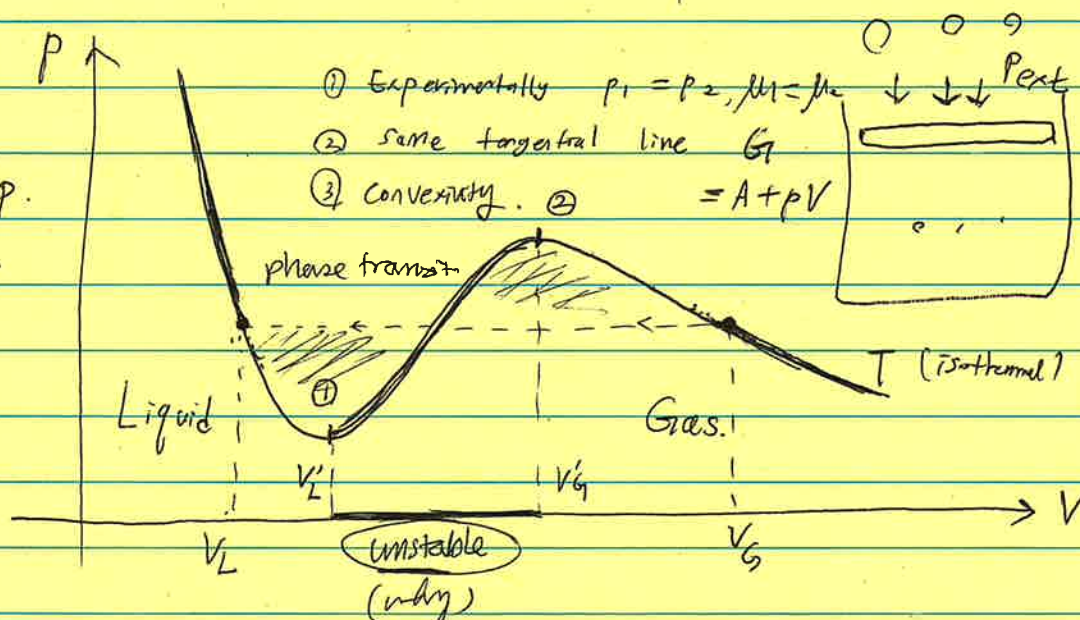
$\mu_s = \mu_B$ recovers. (*) from Free energy minimization.

Free Energy Minimization $\equiv$ Chemical potential equilization, $\#$

Phase Transitions.



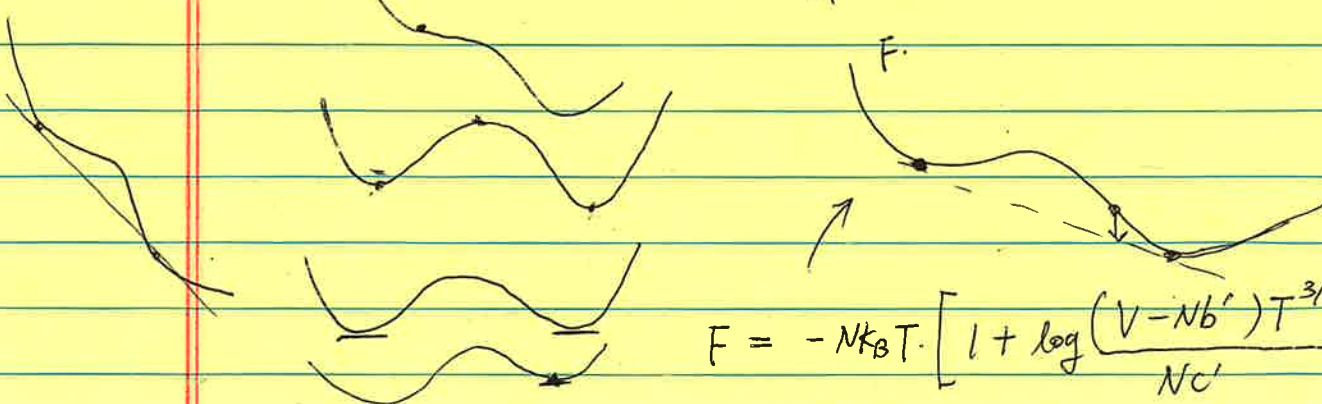
$$\left. \frac{\partial A}{\partial V} \right|_T = -P.$$



- ① Experimentally $P_1 = P_2, \mu_1 = \mu_2$
- ② same tangential line G
- ③ Convexity. ② $= A + pV$

In region ① ②, in a cylinder, decreasing P_{ext} slightly will introduce increase in volume. Such volume increase (according to graph ① ②), will increase the pressure. Then, again, volume increases. Repeats $\rightarrow$ Unstable.

$\Rightarrow$ Maxwell's construction $\left. \frac{\partial P}{\partial V} \right|_T < 0$ (necessary for stability)



$$F = -Nk_B T \left[1 + \log \left(\frac{(V - Nb') T^{3/2}}{Nc'} \right) \right] - \frac{a' N^2}{V}$$

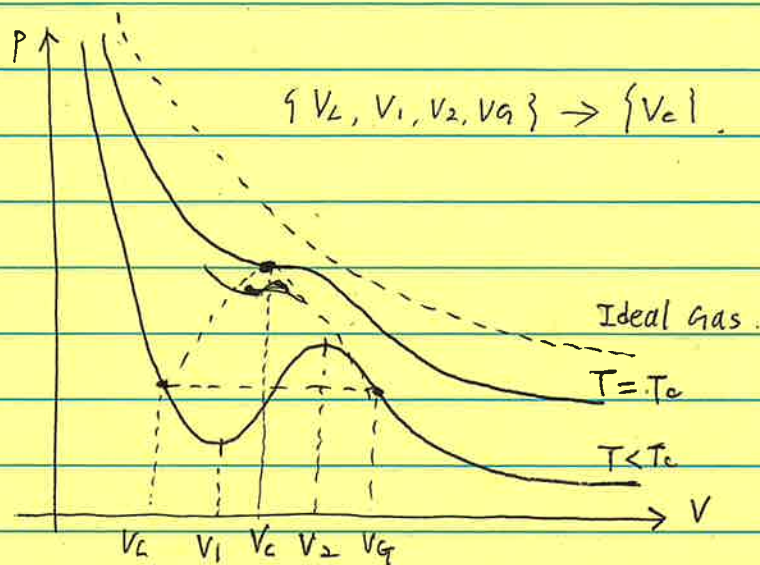
$$\beta p = p$$

- Critical Point of Van der Waals Model.

$$p = \frac{k_B T}{V-b} - \frac{a}{V^2} \quad (*)$$

(At $T = T_c$).

$$\begin{aligned} ① \left. \frac{\partial p}{\partial V} \right|_{V_c} &= 0 \\ ② \left. \frac{\partial^2 p}{\partial V^2} \right|_{V_c} &= 0 \end{aligned} \quad \left. \begin{array}{l} \text{Inflection} \\ \text{point!} \end{array} \right\}$$



$$① \frac{\partial p}{\partial V} = -k_B T \cdot (V_c - b)^{-2} + 2a V_c^{-3} = 0$$

$$\Rightarrow 2a V_c^{-3} = k_B T_c (V_c - b)^{-2}$$

$$② \frac{\partial^2 p}{\partial V^2} = 2k_B T (V_c - b)^{-3} - 6a V_c^{-4} = 0$$

$$\Rightarrow 6a V_c^{-4} = 2k_B T_c (V_c - b)^{-3}$$

$$\Rightarrow \frac{1}{3} \cdot V_c^{-1} = \frac{1}{2} \cdot (V_c - b)^{-1} \Rightarrow 3V_c - 3b = 2V_c \Rightarrow \underline{V_c = 3b}$$

$$(*) \Rightarrow p_c = \frac{k_B T_c}{2b} - \frac{a}{9b^2}$$

$$① \Rightarrow 2a (3b)^{-3} = k_B T_c \cdot (2b)^{-2} \Rightarrow \underline{k_B T_c = \frac{8a}{27b}}, \quad \underline{p_c = \frac{a}{27b^2}}$$

- Non-dimensional variables

$$\hat{p} = p/p_c, \quad \hat{V} = V/V_c, \quad \hat{T} = T/T_c.$$

$$\Rightarrow \hat{p} \cdot p_c = \frac{k_B \hat{T} \cdot T_c}{\hat{V} \cdot V_c - b} - \frac{a}{(\hat{V} \cdot V_c)^2}$$

$$\Rightarrow \hat{p} \cdot \frac{a}{27b^2} = \frac{\frac{8a}{27b} \hat{T}}{3b\hat{V} - b} - \frac{a}{(3b\hat{V})^2}$$

$$\frac{\hat{p}}{27} = \frac{8\hat{T}/27}{3\hat{V} - 1} - \frac{1}{9\hat{V}^2}$$

$$\Rightarrow \left(\hat{p} + \frac{3}{\hat{V}^2} \right) (3\hat{V} - 1) = 8\hat{T}$$

(*)'

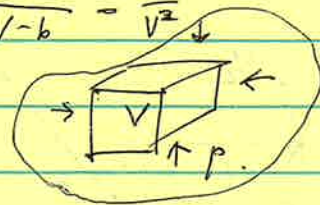
• Critical Exponent (K_T)

Given $\left(\hat{p} + \frac{3}{\hat{V}^2}\right) \cdot (3\hat{V} - 1) = 8\hat{T} \Leftrightarrow p = \frac{k_B T}{V-b} - \frac{a}{V^2}$

Q1) what happens $\hat{V} \approx 1/3 \Leftrightarrow V \approx \frac{1}{3} V_c$

Q2) Isothermal compressibility $K_T = -\frac{1}{V} \frac{\partial V}{\partial p}$

$K_T = -\frac{1}{V} \cdot \frac{\partial V}{\partial p}$



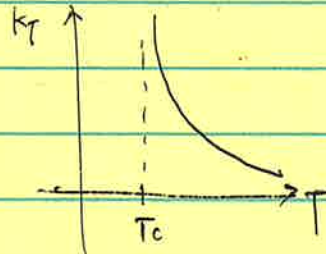
$\rightarrow$ Watch $\partial p / \partial V = -k_B T \cdot (V-b)^{-2} + 2aV^{-3}$

$\rightarrow \partial V / \partial p = \frac{1}{-k_B T \cdot (V-b)^{-2} + 2aV^{-3}} \Rightarrow K_T = \frac{1}{k_B T \cdot V(V-b)^{-2} - 2aV^{-2}}$

$\rightarrow$ Around $T = T_c, V = V_c = 3b$

$K_T = \frac{1}{k_B T \cdot 3b(2b)^{-2} - 2a(3b)^{-2}} \quad \left(\text{and apply } a = \frac{27}{8} b k_B T_c\right)$

$\Rightarrow K_T = \frac{1}{\frac{3k_B}{4b} T - \frac{3k_B}{4b} T_c} = \frac{4b}{3k_B} (T - T_c)^{-1}$
 $\boxed{\gamma = 1}$



• Critical Exponent (V) $T \rightarrow T_c$

Q) How does volume deviate around T_c ?

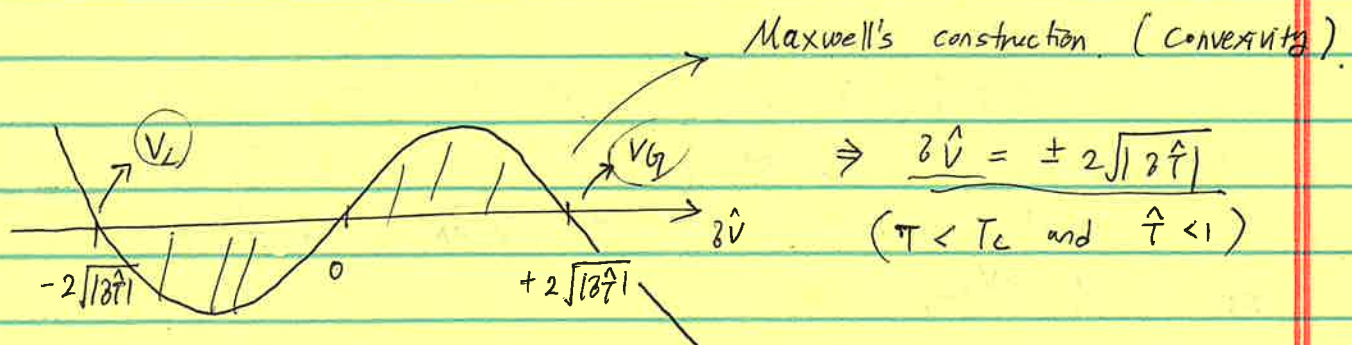
We know non-dimensional relation.

$\hat{p} = \frac{8\hat{T}}{3\hat{V} - 1} - \frac{3}{\hat{V}^2} \quad \left\{ \begin{array}{l} \hat{T} \rightarrow 1 + \delta\hat{T} \\ \hat{V} \rightarrow 1 + \delta\hat{V} \end{array} \right.$

$\hat{p} = \frac{8(1 + \delta\hat{T})}{2 + 3\delta\hat{V}} - \frac{3}{(1 + \delta\hat{V})^2} = \frac{4(1 + \delta\hat{T})}{1 + \frac{3}{2}\delta\hat{V}} - \frac{3}{(1 + \delta\hat{V})^2}$

$\approx 1 + 4\delta\hat{T}$

$\Rightarrow \hat{p} \approx 1 + 4\delta\hat{T} - 6(\delta\hat{T})(\delta\hat{V}) \left(1 + \frac{(\delta\hat{V})^2}{4\delta\hat{T}}\right)$



$$\Rightarrow \hat{v} = 1 \pm 2\sqrt{1 - \hat{f}} \Rightarrow \hat{v} = v/v_c = 1 \pm \sqrt{1 - T/T_c}$$

$$\Rightarrow v_G - v_L = 4v_c \left(1 - T/T_c\right)^{1/2} = 12b \cdot \left(1 - T/T_c\right)^{1/2}$$

Thms, $v_G - v_L$ goes to zero following $\propto (T_c - T)^{1/2}$
 $= (T_c - T)^{\tilde{\beta}}$

$$\tilde{\beta} = 1/2$$

• Problem Session (ME346A)

03/08/2025

• Monte Carlo Simulation (Rosenbluth, Teller, Metropolis)

at Markov chain.

- ① Initial state v
- ② Generate random state v'
- ③ Evaluate probability

$$\frac{p(v')}{p(v)} = \frac{\frac{1}{Z} \cdot e^{-\beta E(v')}}{\frac{1}{Z} \cdot e^{-\beta E(v)}} = e^{-\beta [E(v') - E(v)]} = e^{-\beta \Delta E}$$
- ④ Accept / reject with probability,

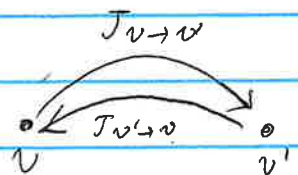
$$p_{acc}(v \rightarrow v') = \min(1, e^{-\beta \Delta E})$$

• Detailed Balance.

$$p(v) \cdot p(v \rightarrow v') = p(v') \cdot p(v' \rightarrow v)$$

$$\Leftrightarrow \frac{p(v \rightarrow v')}{p(v' \rightarrow v)} = \frac{p(v')}{p(v)}$$

Note: $\underbrace{p(v)} \cdot \underbrace{p(v \rightarrow v')} \approx \text{"flux"}$



If you select any pairs of v and v' , (v, v')

$$\text{Fluxes } J_{v \rightarrow v'} = J_{v' \rightarrow v}$$

Global detailed balance $\rightarrow$ No net flow

No net flow $\rightarrow$ Reversible.

Reversible $\rightarrow$ Equilibrium distribution is sampled.

- Examine detailed balance for MCMC.

$$p(v \rightarrow v') = \underbrace{p_{\text{gen}}(v \rightarrow v')}_{\text{probability to generate such move}} \cdot \underbrace{p_{\text{acc}}(v \rightarrow v')}_{\text{probability to accept such move}}$$

From detailed balance,

$$\frac{p(v)}{p(v')} = \frac{p(v' \rightarrow v)}{p(v \rightarrow v')} = \frac{p_{\text{gen}}(v' \rightarrow v) \cdot p_{\text{acc}}(v' \rightarrow v)}{p_{\text{gen}}(v \rightarrow v') \cdot p_{\text{acc}}(v \rightarrow v')}$$

when generating (proposing) probabilities are "symmetric", (i.e. $p_{\text{gen}}(v \rightarrow v') = p_{\text{gen}}(v' \rightarrow v)$)

$$\frac{p(v)}{p(v')} = \frac{p_{\text{acc}}(v' \rightarrow v)}{p_{\text{acc}}(v \rightarrow v')}$$

Metropolis-Hastings.

Apply Monte Carlo, where

$$p_{\text{acc}}(v \rightarrow v') = \min[1, e^{-\beta \Delta E}] \quad (\Delta E = E(v') - E(v))$$

$$\frac{p(v' \rightarrow v)}{p(v \rightarrow v')} = \frac{p(v)}{p(v')} = \frac{\min[1, e^{+\beta \Delta E}]}{\min[1, e^{-\beta \Delta E}]} \stackrel{\text{should be,}}{=} \frac{e^{-\beta E(v)}}{e^{-\beta E(v')}} = e^{+\beta (E(v') - E(v))} = e^{\beta \Delta E}$$

Condition for D.B.

Case 1) $\Delta E > 0$

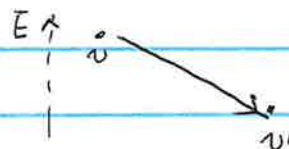


$$\min[1, e^{-\beta \Delta E}] = e^{-\beta \Delta E}$$

$$\min[1, e^{+\beta \Delta E}] = 1$$

$$p(v)/p(v') = e^{+\beta \Delta E}$$

Case 2) $\Delta E < 0$



$$\min[1, e^{-\beta \Delta E}] = 1$$

$$\min[1, e^{+\beta \Delta E}] = e^{+\beta \Delta E}$$

$$p(v)/p(v') = e^{\beta \Delta E}$$

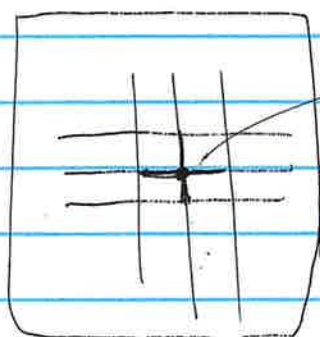
Some notes on Monte Carlo.

① How do you implement it?

Main question: How to probability

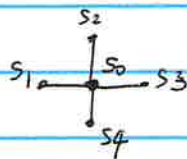
EX) $\text{np.random.rand()} < 0.3$

② How to calculate energy difference in Ising Model?



(i, j)

$$H = -J \sum_{\langle i, j \rangle} s_i s_j - h \sum_i s_i$$



Before: $-h s_0 - h s_1 - \dots - h s_4$

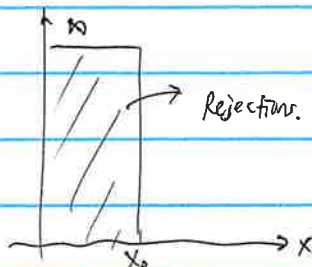
After: $+h s_0 - h s_1 - \dots - h s_4$

Before: $-J s_0 s_1 - J s_0 s_3 - J s_0 s_2 - J s_0 s_4$

After: $-J (-s_0) s_1 - J (-s_0) s_3 - J (-s_0) s_2 - J (-s_0) s_4$

$$\begin{aligned} \Delta H &= -h(-s_0 - s_0) - J(s_1 + \dots + s_4) \cdot (-s_0 - s_0) \\ &= 2 \left(-h \cdot s_0 - J \cdot s_0 (s_1 + \dots + s_4) \right) \end{aligned}$$

③ Hard disk model



[1] Equation of State Calculations by Fast Computing Machines. (1953)

$$F_i = m \cdot \ddot{q}_i \quad (1687)$$

$$\frac{dP_i}{dt} = F_i \quad \text{--- (1688)}$$

$$L = K - U = \sum_i \frac{1}{2} m (\dot{q}_i)^2 - U(q_i)$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0 \quad \text{--- (1689)}$$

$$\frac{d}{dt} (m \dot{q}_i) + \left(+ \frac{\partial}{\partial q_i} U(q_i) \right) = 0$$

$$\Rightarrow \ddot{q}_i = - \frac{1}{m} \nabla U(q_i)$$

$$\ddot{q}_i = - \left(\frac{1}{m} \frac{\partial U}{\partial q_i} \right) \sim 0$$

$$\frac{dL}{dt} = \sum_i \underbrace{\frac{\partial L}{\partial q_i} \frac{dq_i}{dt}}_{(1)} + \underbrace{\frac{\partial L}{\partial \dot{q}_i} \frac{d\dot{q}_i}{dt}}_{(2)} = \frac{dL}{dq_i} \frac{dq_i}{dt} + \frac{dL}{d\dot{q}_i} \frac{d\dot{q}_i}{dt}$$

$$(1) = \frac{\partial L}{\partial q_i} \frac{dq_i}{dt} = \left(\frac{\partial q_i}{\partial q_i} \right) \frac{\partial L}{\partial q_i} \left(\frac{dq_i}{dt} \right) = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) \dot{q}_i$$

$$\Rightarrow \frac{d}{dt} \left[\frac{\partial L}{\partial \dot{q}_i} \dot{q}_i \right] + \frac{dL}{dq_i} \frac{dq_i}{dt} = 0$$

$$(uv)' = u'v + uv'$$

$$\Rightarrow \frac{dL}{dt} = \frac{d}{dt} \left[\underbrace{\frac{\partial L}{\partial \dot{q}_i}}_u \underbrace{\dot{q}_i}_v \right] \Rightarrow \frac{d}{dt} \left[\underbrace{-L + \frac{\partial L}{\partial \dot{q}_i} \dot{q}_i}_H \right] = 0$$

$$H = -L + \frac{\partial L}{\partial \dot{q}_i} \dot{q}_i = -L + p_i \dot{q}_i$$

$$\text{Ex) } H = \frac{p^2}{2m} + U(x) \quad H(x, p)$$

$$p = mv$$

$$L = -H + \left(\frac{\partial H}{\partial p} \right) p = -H + v \cdot p$$

$$① dH = \frac{\partial H}{\partial p} dp + \frac{\partial H}{\partial x} dx = v dp + U'(x) dx$$

$$② dL = -dH + v dp + p dv = -v dp - U'(x) dx + v dp + p dv$$

$$L(x, v)$$

$$H = -L + \left(\frac{\partial L}{\partial v} \right) v$$

Quiz

01/17/2025

If RVs X, Y satisfy $\langle XY \rangle = \langle X \rangle \langle Y \rangle$, they are independent.

0 0

F

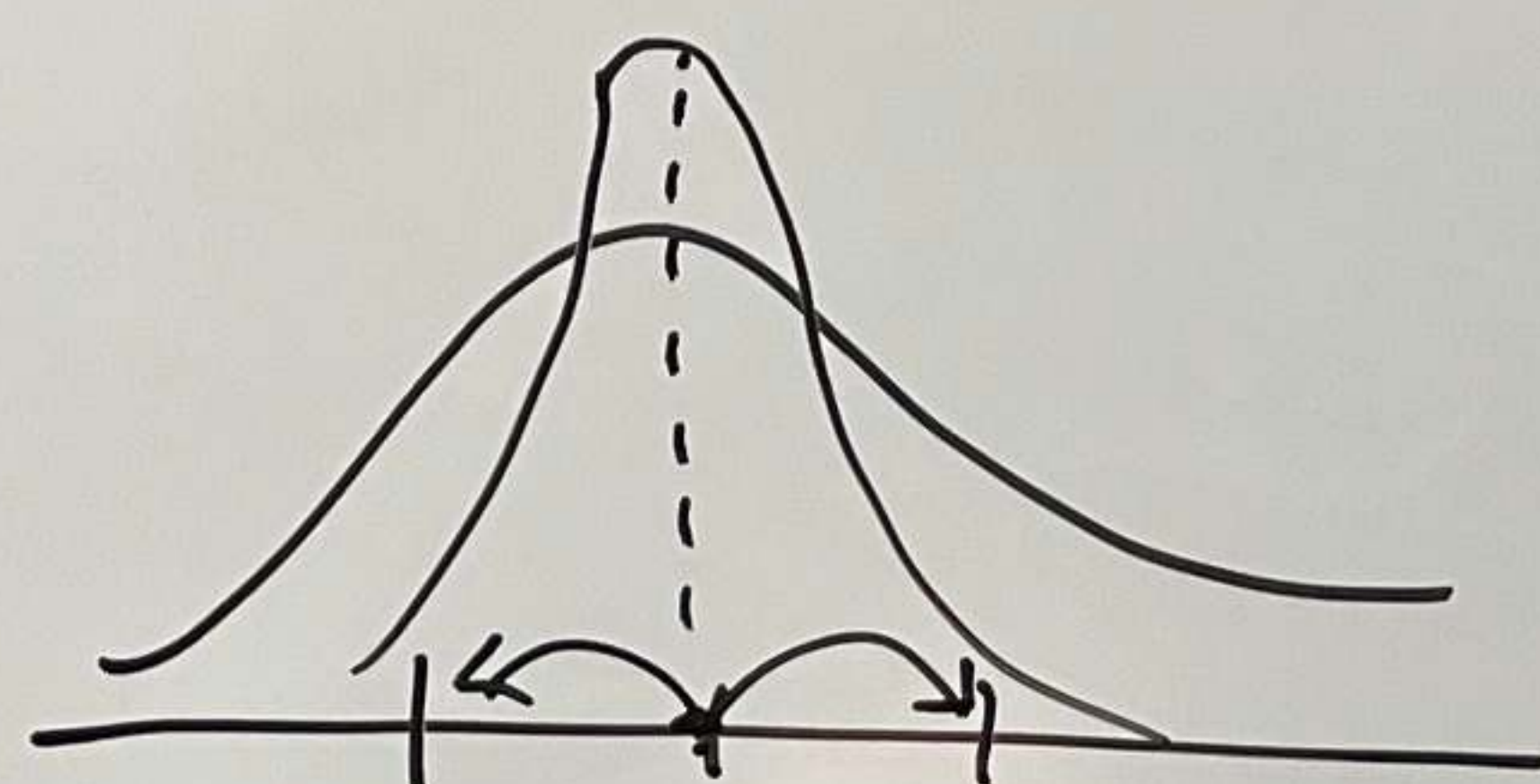
$J = \frac{N_{\text{net}}}{\text{Area} \cdot \text{Time}}$

T

$l_i = \pm 1$ prob: $p = 1/2$
0 prob: $1-2p$

$l_i = \pm 1$ prob: $1/2$

F



$Y = X^2$

$X = [-1, +1]$

$P(Y=y | X=x) = P(Y=y)$

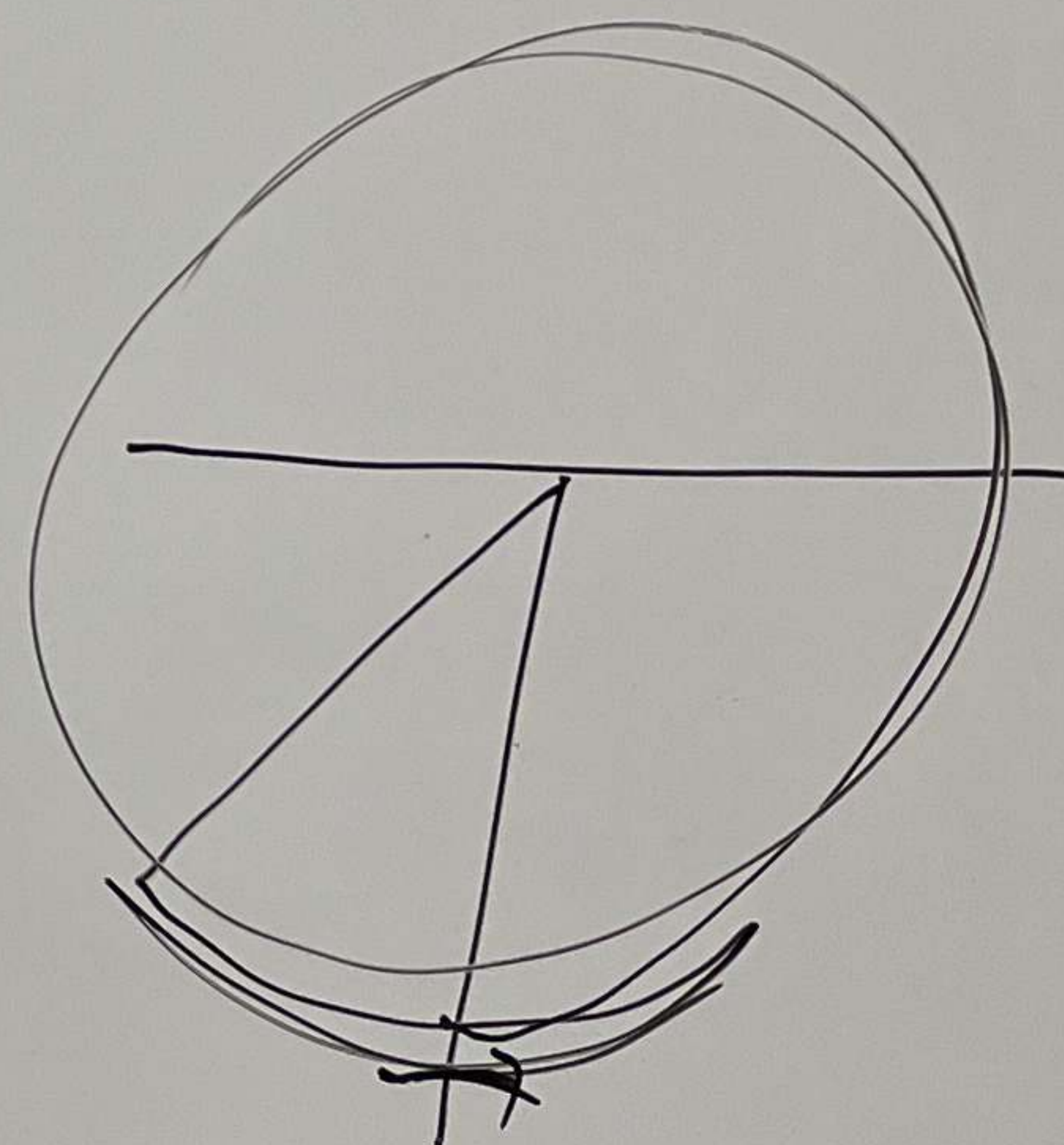
$L = K - U$, L is conserved.

F

U is conservative field. $dH/dt = 0 \Rightarrow$ Conservation of energy.

T

▽



$$F_i = m \cdot \ddot{q}_i \quad (1687)$$

$$\frac{dp_i}{dt} = F_i \quad \text{--- a}$$

$$L = K - U = \sum_i \frac{1}{2} m (\dot{q}_i)^2 - U(q_i) - i(q_i) q_i$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0 \quad \text{--- b} \quad (1760)$$

$$\frac{d}{dt} (m \dot{q}_i) + \left(+ \frac{\partial}{\partial q_i} U(q_i) \right) = 0$$

$$\Rightarrow \ddot{q}_i = - \frac{1}{m} \nabla U(q_i)$$

$$\ddot{q}_i = - \left(\frac{1}{m} \frac{\partial U}{\partial q_i} \right) \sim \text{--- c}$$

$$\frac{dL}{dt} = \sum_i \underbrace{\frac{\partial L}{\partial \dot{q}_i} \frac{d\dot{q}_i}{dt}}_{(1)} + \underbrace{\frac{\partial L}{\partial q_i} \frac{dq_i}{dt}}_{(2)} \quad \text{LHS}$$

$$(1) = \sum_i \frac{\partial L}{\partial \dot{q}_i} \frac{d\dot{q}_i}{dt} = \sum_i \left(\frac{\partial \dot{q}_i}{\partial \dot{q}_i} \right) \frac{\partial L}{\partial \dot{q}_i} \left(\frac{d\dot{q}_i}{dt} \right) \dot{q}_i$$

$$= \sum_i \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) \dot{q}_i$$

$$\Rightarrow \frac{d}{dt} \left[\underbrace{\frac{\partial L}{\partial \dot{q}_i}}_{(1)} \dot{q}_i \right] + \underbrace{\frac{\partial L}{\partial q_i} \frac{dq_i}{dt}}_{(2)} \quad \text{RHS}$$

$$\Rightarrow \frac{dL}{dt} = \sum_i \frac{d}{dt} \left[\underbrace{\frac{\partial L}{\partial \dot{q}_i}}_{\text{LHS}} \underbrace{\dot{q}_i}_{\text{RHS}} \right] \Rightarrow \frac{d}{dt} \left[\underbrace{-L + \sum_i \frac{\partial L}{\partial \dot{q}_i} \dot{q}_i}_{H} \right] = 0$$

$$\begin{cases} \frac{\partial L}{\partial \dot{q}_i} = p_i & \frac{\partial L}{\partial q_i} = \left(\frac{\partial \dot{q}_i}{\partial q_i} \right) \frac{\partial L}{\partial \dot{q}_i} = \frac{d}{dt} p_i = \dot{p}_i \\ \frac{\partial L}{\partial q_i} = \dot{p}_i & \text{EOM L} \end{cases}$$

$$H = -L + p \dot{q}$$

$$dH = dp \dot{q} + \cancel{d\dot{q} p} - \underbrace{\frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i}_{p dq} - \underbrace{\frac{\partial L}{\partial q_i} dq}_{\dot{p} dq} = \underbrace{\dot{q} dp}_{\frac{\partial H}{\partial p}} - \underbrace{\dot{p} dq}_{\frac{\partial H}{\partial q}}$$

$$\text{① } L \leftrightarrow H$$

$$\text{② } dH/dt = 0$$

$$e^{-\beta U}$$

$$U \in C^\infty$$

$$\text{EOM H}$$

$$\begin{cases} \frac{\partial H}{\partial p} = \dot{q} \\ \frac{\partial H}{\partial q} = -\dot{p} \end{cases}$$

$$S = k_B N \left[\log \left(\frac{V}{N} \left(\frac{4\pi m E}{3} \right)^{3/2} \right) + \frac{5}{2} \right]$$

$$(a) E = \frac{3 N k_B}{4\pi} \left(\frac{N}{V} \right)^{2/3} \exp \left[\frac{2S}{3k_B} - \frac{5}{2} \right]$$

$$T = \left(\frac{\partial E}{\partial S} \right)_{N,V} \Rightarrow T = \frac{2}{3 N k_B} E \Rightarrow E = \frac{3}{2} N k_B T$$

$$P = - \left(\frac{\partial E}{\partial V} \right)_{S,N} \Rightarrow P = \frac{2}{3} \frac{1}{V} E \Rightarrow PV = \frac{2}{3} N k_B T$$

$$P = \left(\frac{\partial E}{\partial V} \right)_{S,N} \Rightarrow \frac{2}{3} \frac{1}{V} E$$

S.T. of
Sackur-Tetrode
 $S(N, V, E)$

$$(b) A = E - TS$$

$$= \frac{3}{2} N k_B T - T \left[N \log \left(\frac{V}{N} \left(\frac{4\pi m k_B T}{3} \right)^{3/2} \right) + \frac{5}{2} \right] k_B$$

$$(c) G = E - TS + PV$$

$$= - N k_B T \left[\log \left(\frac{V}{N} \left(\frac{2\pi m k_B T}{3} \right)^{3/2} \right) \right] (N, V, T)$$

= ?

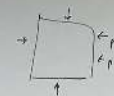
$$(d) \frac{\partial N}{\partial T} = T \left(\frac{\partial S}{\partial T} \right)_{N,V} = \frac{3}{2} N k_B \quad (N, V, T)$$

$$C_V = \frac{\partial Q}{\partial T} = T \left(\frac{\partial S}{\partial T} \right)_{N,V} = \frac{3}{2} N k_B \quad (N, V, T)$$

$$C_P - C_V = N k_B$$



$$(e) \alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_{E,N} = \frac{1}{V} \frac{\partial}{\partial T} \frac{N k_B T}{P} = \frac{1}{V} \frac{N k_B}{P} = \frac{1}{P}$$



$$\beta = - \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_{E,N} = \frac{1}{V} \frac{N k_B T}{P} = \frac{1}{P}$$

$$\frac{\partial A}{\partial T} = -S - T \frac{\partial S}{\partial T} = -S$$

$$Z = \sum_v e^{-\beta E(v)} \approx \int dE \Omega(E) e^{-\beta E(v)} \quad \text{Laplace}$$

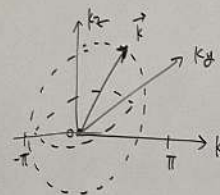
$$= \int dE e^{-\beta \left[\underbrace{E(v) - \log \Omega(E)}_{A(v)} \right]} \quad \text{Legendre}$$

Ω	E
Z	A
E	G
Δ	$?$
TP	

Density of States.

$\vec{k}$: wave vector.

$$\vec{k} = \left(\frac{2\pi}{L}\right)(n_x, n_y, n_z)$$

$$\frac{2\pi}{L} \cdot n = \frac{2\pi}{\lambda}$$


$$g(k)dk = \frac{4\pi k^2 dk}{(2\pi/L)^3} = \frac{L^3}{2\pi^2} k^2$$

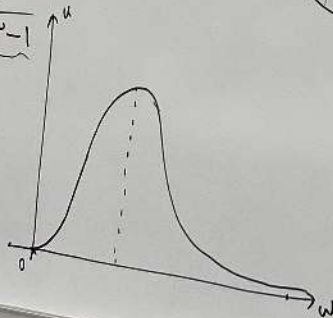
Planck's law $\mu=0$

$$u \propto \int \epsilon(\omega) \cdot g(\omega) d\omega \cdot \langle n_s \rangle_B = \frac{1}{e^{\beta \hbar \omega} - 1} \cdot \omega \cdot \omega^2 d\omega$$

$$\propto \frac{\omega^3}{e^{\beta \hbar \omega} - 1}$$

$\omega \rightarrow 0 \quad u \propto \omega^2 \rightarrow \text{Rayleigh-Jeans law}$

$\omega \rightarrow \infty \quad u \rightarrow 0$



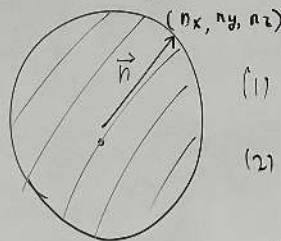
photons

$\odot \uparrow \downarrow$ $\odot \sim \sim \sim$ $\S$: polarization.

$$\omega = 2\pi f$$

$$g(\omega) d\omega = \frac{\partial N(\omega)}{\partial \omega} \cdot d\omega$$


$$g(k) dk = \frac{\partial N(k)}{\partial k} dk$$



$$(1) N = \frac{4}{3} \pi \|\vec{k}\|^3 \propto \|\vec{p}\|^3 \propto \|\vec{\epsilon}\|^3 \propto \omega^3 \rightarrow g(\omega) d\omega \propto \omega^2 d\omega$$

$$(2) 4\pi \|\vec{k}\|^2 \cdot d\|\vec{k}\| \propto \omega^2 d\omega = g(\omega) d\omega$$

$$g(\omega) d\omega \propto \omega^2 d\omega$$

$$p = \frac{h}{\lambda} = \frac{hf}{c} \Rightarrow pc = \epsilon$$

$$k = \frac{2\pi}{\lambda} \quad \epsilon = hf = \hbar \omega$$

$$p = \hbar k$$

$$\|\vec{p}\| \propto \hbar \|\vec{n}\|$$

$$\Rightarrow \|\vec{n}\| \propto \|\vec{p}\|$$

Grand Canonical Ensemble.

$$\mathcal{Z}_{\text{Boson}} = \sum_{n_0=0}^{\infty} \sum_{n_1=0}^{\infty} \dots \sum_{n_M=0}^{\infty} e^{-\beta (\sum_j n_j \epsilon_j - \mu \sum_j n_j)}$$

$$= \prod_{j=0}^N \sum_{n_j=0}^{\infty} e^{-\beta (\epsilon_j - \mu) n_j} \quad \left\{ \sum_{n=0}^{\infty} r^n = \frac{1}{1-r} \right\}$$

$\rightarrow \frac{1}{1 - e^{-\beta(\epsilon_j - \mu)}}$

$-\beta(\epsilon_j - \mu) < 0 \rightarrow \epsilon_j > \mu$

$$\mathcal{Z}_B = \prod_{j=0}^N \frac{1}{1 - e^{-\beta(\mu + \epsilon_j)}}$$

$$\mathcal{Z}_F = \prod_{j=0}^N (1 + e^{-\beta(\epsilon_j - \mu)})$$

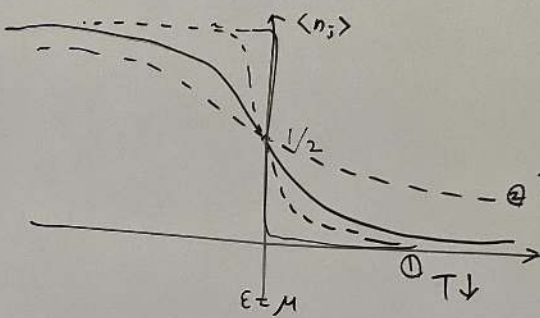
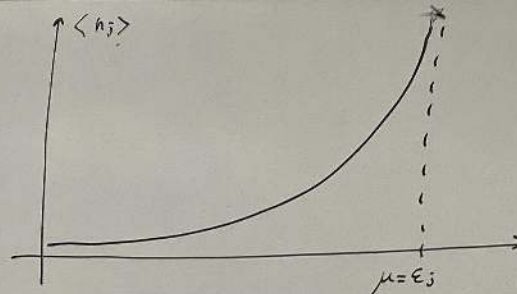
$$\langle n_j \rangle_{\text{Bosons}} = \frac{\partial \log \mathcal{Z}_B}{\partial (-\beta(\epsilon_j - \mu))} = \frac{1}{e^{\beta(\epsilon_j - \mu)} - 1}$$

$$\langle n_j \rangle_{\text{Fermion}} = \frac{1}{e^{\beta(\epsilon_j - \mu)} + 1}$$

$$\rightarrow \frac{\partial \log \mathcal{Z}}{\partial (-\beta)} = \frac{1}{\mathcal{Z}} \frac{\partial \mathcal{Z}}{\partial (-\beta)}$$

$$\Delta n_{j \text{ Bosons}} = \frac{\partial^2 \log \mathcal{Z}_B}{\partial (-\beta(\epsilon_j - \mu))^2}$$

$$\Delta n_{j \text{ F.}} \quad \beta = \frac{1}{k_B T}$$



critical phenomena

$$\left. \begin{array}{l} \hat{p} = p/p_c \\ \hat{V} = V/V_c \\ \hat{T} = T/T_c \end{array} \right\} \rightarrow \left(\hat{p} + \frac{3}{\hat{V}^2} \right) \cdot \left(3\hat{V} - 1 \right) = 8\hat{T}$$

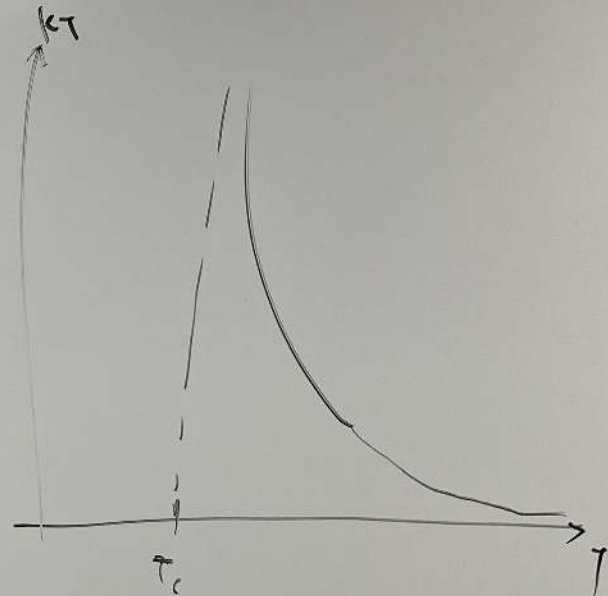
$$k_T = -\frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_T$$

$$-\frac{\partial p}{\partial V} = -k_B T (V-b)^{-2} + 2aV^{-3}$$

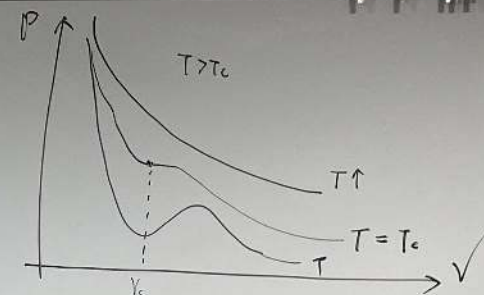
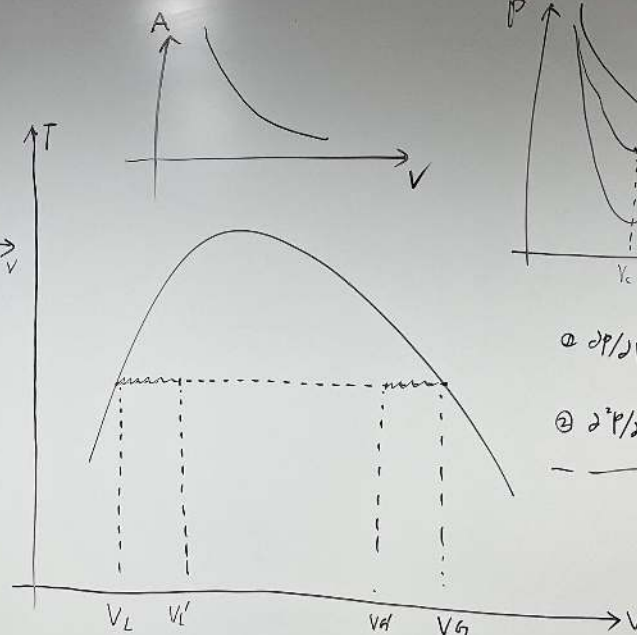
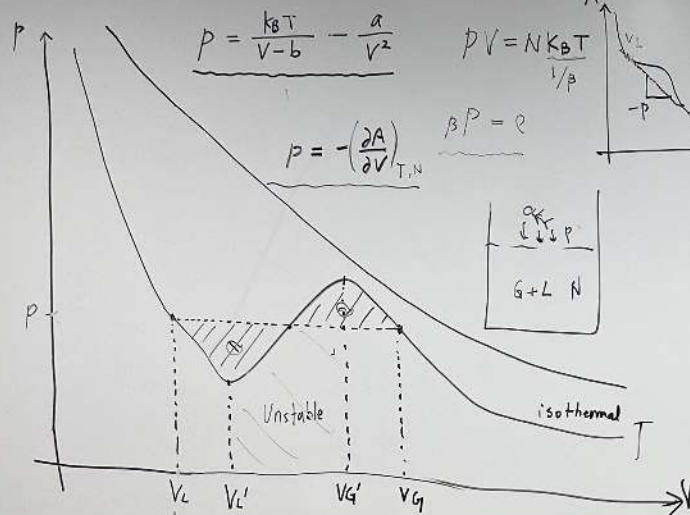
$$\frac{\partial V}{\partial p} = \frac{1}{-k_B T (V-b)^{-2} + 2aV^{-3}}$$

$$k_T = -\frac{1}{V} \frac{\partial V}{\partial p} = \frac{1}{k_B T \cdot V (V-b)^{-2} - 2aV^{-2}}$$

$$= \frac{4b}{3k_B} (T - T_c)^{-1} \quad T > T_c \quad \alpha = 1$$



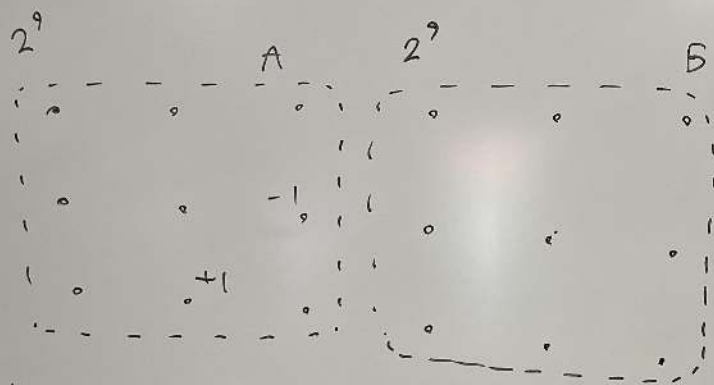
• Van der Waals Model



$$\textcircled{1} \frac{\partial P}{\partial V} \Big|_{V_c} = 0 \Rightarrow 2aV_c^{-3} = k_B T_c (V_c - b)^{-2}$$

$$\textcircled{2} \frac{\partial^2 P}{\partial V^2} \Big|_{V_c} = 0 \Rightarrow 6aV_c^{-4} = 2k_B T_c (V_c - b)^{-3}$$

$$\boxed{\begin{aligned} V_c &= 3b \\ k_B T_c &= \frac{8a}{27b} \\ P_c &= \frac{a}{27b^2} \end{aligned}}$$



$$18 \text{ spins} = 2^{18}$$

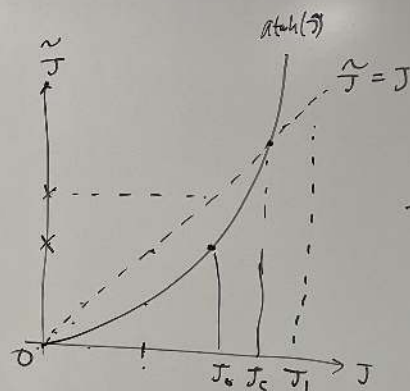
$$\sigma_A = \begin{cases} +1 & \sum_{i,j \in A} > 0 \\ -1 & < 0 \end{cases}$$

$$Z = \sum_{\sigma_A, \sigma_B} \tilde{Z}(\sigma_A, \sigma_B)$$

$$\tilde{Z} = e^{-\beta \tilde{H}} \quad \tilde{H} = -\tilde{J} \sigma_A \sigma_B - C - \tilde{h} \sigma_A - \tilde{h} \sigma_B$$

MATH.

$$\beta \tilde{J} = \text{arctanh}(\beta J)$$



$$\frac{Z_{+1,+1} - Z_{+1,-1}}{Z_{+1,+1} + Z_{+1,-1}} = \frac{e^{\beta \tilde{J} + \beta h} - e^{-\beta \tilde{J} - \beta h}}{e^{\beta \tilde{J} + \beta h} + e^{-\beta \tilde{J} - \beta h}}$$

$$= \tanh\left(\frac{\beta \tilde{J} + \beta h}{\tilde{J}}\right) = \frac{\beta J}{\beta \tilde{J}}$$

$$\begin{pmatrix} \tilde{Z}_{+1,+1} & \tilde{Z}_{+1,-1} \\ \tilde{Z}_{-1,+1} & \tilde{Z}_{-1,-1} \end{pmatrix} = e^{\beta C} \begin{pmatrix} e^{\beta \tilde{J} + \beta h} & e^{-\beta \tilde{J}} \\ e^{-\beta \tilde{J}} & e^{\beta \tilde{J} - \beta h} \end{pmatrix}$$

• Monte Carlo

- ① v
- ② v' } ΔE
- ③ $\frac{p(v')}{p(v)} = \frac{e^{-\beta E(v')}}{e^{-\beta E(v)}} = e^{-\beta(E(v') - E(v))} = e^{-\beta \Delta E}$
- ④ Accept/Reject

$$P_{\text{acc}}(v \rightarrow v') = \min[1, e^{-\beta \Delta E}] = 0.1$$

Metropolis - Hastings

• Detailed Balance $\equiv$ No net flow

Equilibrium $\leftarrow$ Reversible

$$p(v) \cdot p(v \rightarrow v') = p(v') \cdot p(v' \rightarrow v)$$

$\leftarrow$ Flux

$$P(v \rightarrow v') = \frac{p(v \rightarrow v')}{p_{\text{gen}}(v \rightarrow v')} P_{\text{acc}}(v \rightarrow v')$$

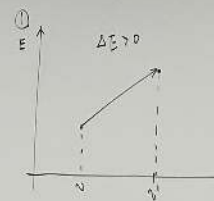
$p_{\text{gen}}(v' \rightarrow v)$ (if symmetric)

Python - How to prob.
 $[0,1]$ $\frac{1}{b}$ $p < 1$
 if $\text{np.random.rand()} \leq p$

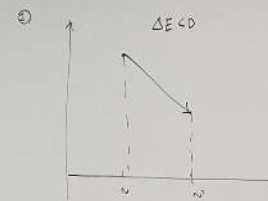


• D.B for M.C.

$$\frac{p(v')}{p(v)} = \frac{p(v \rightarrow v')}{p(v' \rightarrow v)} = \frac{\min(1, e^{-\beta \Delta E})}{\min(1, e^{+\beta \Delta E})}$$

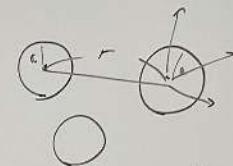
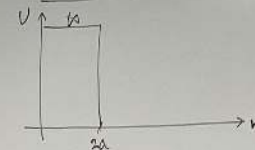


$$\frac{p(v')}{p(v)} = \frac{e^{-\beta \Delta E}}{1} = e^{-\beta \Delta E}$$



$$\frac{p(v')}{p(v)} = \frac{1}{e^{+\beta \Delta E}} = e^{-\beta \Delta E}$$

• Notes



$$|r_{ij} - r_{ij'}|^r$$



PF 3

$$\vec{k} = (k_x, k_y, k_z) = \left(\frac{2\pi}{L}\right) (n_x, n_y, n_z) \quad E_0 = mc^2$$

$$E = (p^2 c^2 + m^2 c^4)^{1/2} \quad \vec{p} = \hbar \vec{k} \quad p^2 c^2 = \hbar^2 k^2 c^2 \quad p^2 = \hbar^2 k^2$$

(a) $g(E)$

(b) $E - E_0 \ll E_0$

(c) $E - E_0 \gg E_0$

$$E^2 - E_0^2 = \hbar^2 k^2 c^2 \quad (1)$$

$$2E dE = \hbar^2 c^2 k dk$$

$$\frac{dE}{dk} = \frac{\hbar^2 c^2}{E} k \rightarrow dk/dE = \frac{E}{\hbar^2 c^2} \frac{1}{k}$$

$$g(E) dE = dN$$

$$g(E) = \frac{dN}{dE} = \frac{dN}{dk} \cdot \frac{dk}{dE}$$



$$\frac{dN}{dk} = \frac{4\pi k^2}{(2\pi/L)^3}$$

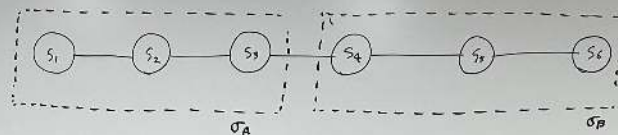
$$g(E) \propto E |E - E_0^2|$$

$$dN = 4\pi k^2 dk \cdot \left(\frac{L}{2\pi}\right)^3$$

$$N = \frac{4}{3} \pi k^3 \cdot \left(\frac{L}{2\pi}\right)^3$$

$$N = \frac{4}{3} \pi n^3 \cdot \left(\frac{L}{2\pi}\right)^3$$

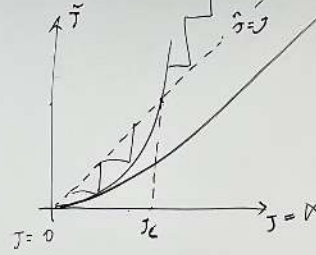
PF 5



$$H = -J \sum_{\langle i,j \rangle} S_i S_j \quad (J > 0)$$

$$\sigma_A = \begin{cases} +1 & S_1 + S_2 + S_3 > 0 \\ -1 & S_1 + S_2 + S_3 < 0 \end{cases}$$

$$\sigma_B = \begin{cases} +1 & S_4 + S_5 + S_6 > 0 \\ -1 & S_4 + S_5 + S_6 < 0 \end{cases}$$



(Q) Find J_c

$$\tilde{H} = -\tilde{J} \sigma_A \sigma_B - C$$

1)	+	+	→ 16	$\sum_{\{S_i\}} z(\sigma_A = +1, \sigma_B = +1)$
	+	-	→ 16	
	-	+	→ 16	$\sum_{\{S_i\}} z$
	-	-	→ 16	

$$\tilde{z}(+,+) = \sum_{\{S_i\}} z(S_1 \sim S_6)$$

$$\tilde{z}(+,-) = \frac{\rho \tilde{J}}{\rho J} ?$$

$$\tilde{J} = f(J)$$

$$J_c$$

$$3 - (b) \quad E - E_0 \ll E_0$$

$$g(E) \propto E \sqrt{E^2 - E_0^2} = E \sqrt{(E - E_0)(E + E_0)} = E \left(1 + \left(\frac{pc}{E_0} \right)^2 \right)^{1/2}$$

$$E = (p^2 c^2 + m^2 c^4)^{1/2} = E_0 \left(1 + \left(\frac{pc}{E_0} \right)^2 \right)^{1/2} = E_0 \left(1 + \frac{1}{2} \left(\frac{pc}{E_0} \right)^2 \right) = E_0 \left(1 + \frac{1}{2} \left(\frac{p}{mc} \right)^2 \right)$$

$$E = E_0 \left(1 + \frac{1}{2} \left(\frac{p}{mc} \right)^2 \right)$$